Topology-Aligned Gated Interaction Network for Cross-Domain Hyperspectral Image Classification

Cuiyu Yang, Lin Gao, Yongxin Feng, and Junchang Xin

***Abstract*—Deep learning algorithms have shown great potential in the field of domain adaptation (DA) in unsupervised learning. However, in the task of cross-scene hyperspectral image (HSI) classification, current DA methods often focus only on local features or semantic information while ignoring the complementary relationship between the source and target domains. Thus, the cross-domain feature discrepancy limits the adaptation performance. To address these issues, we propose a novel topologically aligned gated interaction network (TAGIN) for cross-scene HSI classification. First, we introduce the bidirectional feature interaction network (BFIN), a novel framework in which convolutional neural network (CNN)-extracted local features dynamically interact with Transformer-derived semantic representations through a gating mechanism, forming dual-channel feature information. As a core part of BFIN, a cross-domain spectral space cross-attention (SSCA) mechanism is designed, combined with the bidirectional dynamic feature retention mechanism, to achieve adaptive feature fusion and cross-domain feature selection. Second, to achieve the feature alignment, we design a dynamic adaptive topology alignment loss, which is gradually optimized via memory enhancement and dynamic parameter adjustment mechanisms. Finally, we propose a topology-aware multiview consensus pseudo-label optimization (TA-MVCPO) strategy. High-confidence pseudo-labels are screened out through data augmentation samples, and then, the credibility of the pseudo-labels is verified through topological structure difference perception. The training model was fine-tuned based on high-confidence pseudo-labels, enhancing the adaptability of the training model to the target-domain data. The experimental results conducted on four cross-scenario HSI datasets demonstrate the superior performance of our proposed method.**

***Index Terms*—Hyperspectral image (HSI), image classification, topological alignment, unsupervised domain adaptation (DA).**

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I. Introduction

HYPERSPECTRAL imaging is an imaging technology that combines spectral analysis capabilities and spatial characteristics. The purpose of this combination is to obtain the spectral information of the target object through continuous narrowband acquisition. The technology is capable of providing a continuous spectral curve for each pixel [1]. The primary function of the device is its high-resolution capacity, which facilitates the detection of minute spectral variations [2], [3]. Hyperspectral imaging has a long history in agriculture, environmental monitoring, and other fields, and is highly valued in a range of applications [4], [5], [6]. In recent years, there has been a notable advancement in technology and the development of computers, which has led to a significant trend of integrating remote sensing and artificial intelligence [7], [8]. The fusion of deep learning, reinforcement learning, and hyperspectral images (HSIs) provides the necessary technical foundation for the extraction and classification of spectral features. Feng et al. [9] proposed a novel deep reinforcement learning modeling method for the fixed and adaptive band selection of HSIs. Amoako et al. [10] proposed an HSI classification framework based on meta-reinforcement learning. By integrating the capsule network with deep Q-learning, they achieved excellent classification performance with only a small number of labeled samples. By designing a reward factor to avoid repeated band selection, the flexibility and efficiency of the selection were enhanced. Nowadays, drones equipped with hyperspectral sensors are also typically equipped with other sensors, and multimodal data have become widely applied in numerous fields [11].

Compared with the traditional single-scene HSI classification, the key to cross-scene classification lies in the existence of specific scenes and sufficient label quantities. These factors are of vital importance for achieving satisfactory classification results. However, in reality, the labeled components of HSIs are very high, and there is a phenomenon of insufficient usability [12], [13]. In the context of the rapid development of deep learning, the potential of deep domain adaptive methods in transfer learning is evident. These methods facilitate the transfer of learned knowledge from the source domain (labeled data) to the target domain (unlabeled data) [14]. Presently, it has become a focal point in contemporary research.

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Deep domain adaptive methods has been established under the condition of neural networks and have gradually formed various mature and stable frameworks [15], [16], [17]. Li et al. [18] proposed a loss termed domain separation, which employs neural networks to project the features of the source domain and the target domain onto a unified new domain. The process serves to enhance the similarity between the source and target domains, thereby reducing the disparities between them. Huang et al. [19] proposed a spatial–spectral weighted adversarial domain adaptive (SSWADA) network. The network integrates a domain discriminator, employs a weighted clustering method to weight the source samples, and designs a multiclassifier structure to improve the main classifier. The combination of techniques achieves excellent results in cross-scene HSI wetland classification and mapping. Qiu et al. [20] advanced the concept of domain fusion contrastive learning, a pioneering approach that integrates the contrastive learning of the source domain and the target domain to establish an interaction. The interaction facilitates the identification of analogous samples across domains, thereby enhancing the learning process. Cal et al. [21] integrated the attention mechanism into the neural network, thereby establishing the information interaction between the source domain and the target domain in the feature extraction stage with a view to achieving feature-level alignment. Zhao et al. [22] designed an unbiased randomized network and adopted self-supervised learning and adversarial training methods in this domain to enhance the adaptability of the target domain. Zhang et al. [23] developed a local robust domain adaptive (LRDA) method for cross-domain data recognition. The approach employed statistical alignment to minimize domain transfer between the source domain and the target domain, while integrating the manifold regularization term to facilitate automated learning of the nearest neighbors and weights between the two domains, thereby reducing the disparities between them. In addition, recent studies have also begun to focus on the domain shift problem in the spectral dimension of HSIs. For instance, the shift-sensitive spatial-spectral disentangling learning (S4DL) framework proposed by Feng et al. [24] explicitly separates domain-invariant and domain-specific features through gradient-guided spectral–spatial decoupling and adaptive shift monitoring mechanisms, thereby alleviating the confusion problem between spectral channels.

The abovementioned methods has achieved remarkable results in the field of domain adaptation (DA). However, the majority of these methods use convolutional neural network (CNN) for the extraction of local information, while disregarding the unique representation of Transformer semantic information and lacking interaction and integration between disparate types of features. Consequently, they are unable to establish a complementary relationship, which hinders the provision of accurate information for the recognition and classification of ground objects. In the analysis of interdomain differences, the establishment of a topological structure of similarity differences between the features of the source domain and the target domain, as well as the historical features of the target domain, can effectively broaden the spatial representation of features and adapt to the changes of nonlinear structures. Therefore, following comprehensive consideration of the high-dimensional and complex information of HSIs, the integration of multisource feature information and topological structure analysis is a valuable approach for in-depth study in domain adaptive methods.

In order to resolve the aforementioned issues, this article proposes a topologically aligned gated interaction network (TAGIN). In this network, a bidirectional feature interaction network (BFIN) based on CNN local features and Transformer semantic information was proposed for the first time. The gating mechanism facilitates the interaction between the local features extracted by CNN and the semantic information extracted by Transformer, thereby forming dual-channel feature representation information. Subsequently, the objectives of adaptive feature fusion and cross-domain feature screening are accomplished through the utilization of the cross-domain spectral space cross-attention (SSCA) mechanism and the bidirectional dynamic feature preservation mechanism. In order to achieve feature-level alignment between the source domain and the target domain, a dynamic adaptive topological alignment loss has been designed. The progressive optimization of cross-domain feature alignment has been achieved through the target-domain memory enhancement and dynamic parameter adjustment mechanism. Finally, we designed a topology-aware multiview consensus pseudo-label optimization (TA-MVCPO) strategy. High-confidence pseudo-labels are screened out through data augmentation samples, and then, the credibility of the pseudo-labels is reverified through topological structure difference perception. On this basis, the training model is fine-tuned to enhance its adaptability to the data in the target domain.

The contributions of our proposed TAGIN can be summarized as follows.

1) We propose a novel TAGIN, which embeds a BFIN module in the feature extractor. BFIN integrates CNN-extracted local features dynamically with Transformer-derived semantic representations through an adaptive gating mechanism, thereby establishing a synergistic dual-channel feature fusion framework. Within BFIN, we introduce a novel cross-domain SSCA mechanism coupled with bidirectional dynamic feature retention, which effectively facilitates adaptive feature fusion and cross-domain feature selection.
2) We propose a dynamic adaptive topological alignment loss function that quantifies the similarity discrepancy between source and target domains to establish adaptive topological relationships. By storing historical features in a momentum-updated target-domain memory bank, this approach enhances topological structure continuity across domains while progressively optimizing cross-domain feature alignment.
3) A TA-MVCPO algorithm is designed. Specifically, during the training process, high-confidence pseudo-labels for multiview prediction are screened out through data augmentation samples. Finally, the credibility of the pseudo-label is reverified by the perception of topological structure differences.

This article is organized into five sections. Section II introduces the relevant work. Section III details our proposed method. Section IV presents a comprehensive

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analysis of experimental results. This article concludes in Section V.

II. Related Work

A. Gating Mechanism

The gating mechanism is a key mechanism for neural networks to dynamically adjust their behavior based on input data. The fundamental principle of this mechanism is conditional computation, in which the retention or discarding of information is governed by learnable weights. Typically, a threshold value ranging from 0 to 1 is generated using the sigmoid or softmax function, and then multiplied by the input feature by element. In their pioneering work, Hochreiter and Schmidhuber [25] introduced gating into long short-term memory (LSTM) networks to address the problem of vanishing gradients. In recent years, research methods based on deep learning have become the focus of academic attention. Researchers have gradually integrated gating mechanisms into various neural network architectures. For instance, Hu et al. [26] employed a channel attention mechanism (combining global average pooling with a fully connected layer) to dynamically adjust the channel weights in the squeeze-and-excitation networks. Li et al. [27] proposed the concept of selective kernel networks, which have been shown to effectively fuse multiscale features by utilizing gating mechanisms. Hua et al. [28] replaced traditional attention mechanisms with gated linear units (GLUs) within Transformers to mitigate computational complexity.

In the domain of HSI classification, challenges such as the intricacy of spectral bands and noise interference have significantly impeded the efficacy of feature extraction. In order to remediate these issues, researchers have integrated gating mechanisms into neural networks, with the objective of mitigating these impacts. For instance, Zhong et al. [29] were the first to integrate residual networks into HSI classification by combining 3-D–2-D CNN and gated enhanced spatial–spectral feature fusion techniques. Hong et al. [30] introduced gated multihead attention, a specialized framework designed for the effective filtration of key spectral bands. In summary, the gating mechanism has emerged as a pivotal component in the advancement of image classification tasks [31], [32], [33].

B. Topological Structure Alignment

In the task of unsupervised DA, aligning or establishing the connection between the source domain and the target domain and minimizing the differences are the keys to effectively transferring the knowledge trained in the source domain to the target domain. Early researchers used the maximum mean difference (MMD) and adversarial training to minimize the global statistical difference between the source domain and the target domain, achieving feature alignment [34], [35], [36]. However, these methods only focus on the global distribution and ignore the local spatial–spectral topological relationship of ground objects in HSIs [37]. With the development of related technologies such as deep learning, display modeling topological structures have gradually come into our view. Zhang et al. [38] proposed to align the graph topological structure of the source domain and the target domain through graph optimal transmission, combined with the MMD to achieve collaborative alignment, significantly improving the classification accuracy across scenarios. This team has, for the first time, incorporated local sample relationships (such as adjacency matrices) into the DA framework to address the topological shift problem of similar ground objects in HSIs caused by differences in imaging conditions. Based on the symmetric network framework of multiclass scoring disagreement (MCSD) divergence, Zhang et al. [39] simultaneously optimized the global distribution and local topology, and achieved topology-sensitive domain alignment through the concatenation of dual classifiers, which supported the theoretical rationality of adversarial training. Shang et al. [40] introduced topological structures in joint adversarial networks for the first time. Through adversarial learning at the domain and class levels, they broke through the limitations of traditional methods in local perception.

III. Proposed Method

In the unsupervised domain adaptive (UDA) task, the main objective is to migrate the source-domain data model with sufficient labels to the target domain when the target-domain data do not have real labels. Specifically, the source-domain sample set and the label set, respectively, for $X^s = \{x_1^s, x_2^s, \ldots, x_n^s\}$ and $Y^s = \{y_1^s, y_2^s, \ldots, y_n^s\}$. Target-domain sample set is expressed as $X^t = \{x_1^t, x_2^t, \ldots, x_n^t\}$. The overall workflow diagram of TAGIN is shown in Fig. 1.

In TAGIN, the source domain and the target domain share an encoder E. E mainly consists of two parts: CNN is used to extract local features of HSIs, and Transformer is used to obtain semantic information. Therefore, the CNN is expressed as the source-domain feature extracting x_c^s and the Transformer to extract semantic information is expressed as x_f^s as well as the characteristics of the target domain to x_c^t and x_f^t. The main structure of the encoder E is shown in Fig. 2. In order to fully utilize the local spatial information and semantic information of the data, we have embedded a BFIN in the encoder E. Its purpose is to adaptively screen and fuse the local features and semantic information. In BFIN, the cross-domain SSCA mechanism realizes the interaction information between the features of the source domain and the target domain, thereby generating gating weights. The bidirectional dynamic feature preservation mechanism assesses the importance of features through multiscale consistency verification and the Sinkhorn algorithm. Finally, these two methods are combined to generate mask information for feature screening. In addition, in this module, we optimize the model by calculating the sparse loss $L_{sparsity}$ and the square difference loss L_{mse}. We designed a topological structure loss L_{tal} for the similarity difference between the source domain and the target domain to dynamically and adaptively establish the alignment relationship. This kind of loss not only focuses on the current local structure, but also can establish the global structure between the historical features of the source domain and the target domain, so as to achieve the progressive alignment of cross-domain features more comprehensively. To obtain reliable sample pseudo-labels in the target domain, we adopt the method

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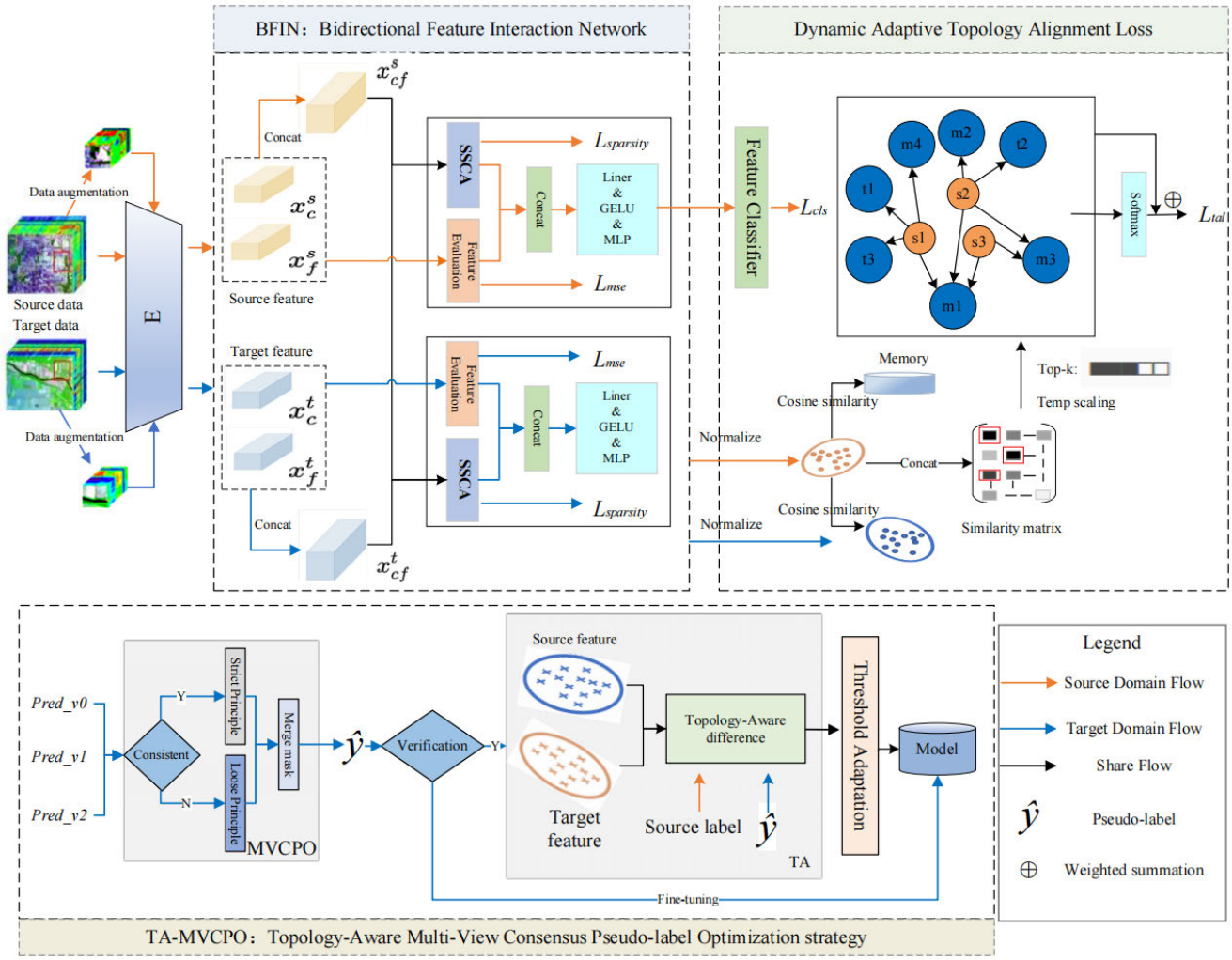


Fig. 1. Overall workflow diagram of TAGIN, where the orange line represents the source-domain process and the blue line represents the target-domain process.

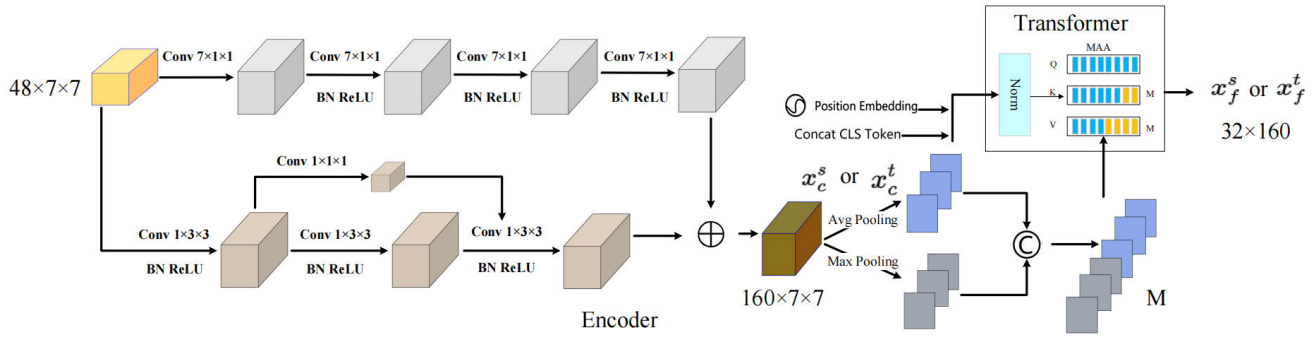


Fig. 2. Overall flowchart of encoder (E) shown in Fig. 1. Taking a sample with input data of $48 \times 7 \times 7$ as an example, CNN is divided into two branches. The upper layer is used to extract spectral features, and the lower layer is used to extract spatial features. Transformer extracts semantic information based on Avg pooling, and Avg pooling and Max pooling are concatenated as the memory enhancement part [41].

of data augmentation. The purpose of data augmentation is to predict the prediction probability of the target domain from multiple perspectives and screen out pseudo-labels with high confidence. Then, the credibility of the pseudo-labels is reverified through the perception of topological structure differences, further increasing the confidence threshold to ensure that high-confidence pseudo-labels are used for model fine-tuning.

A. Bidirectional Feature Interaction Network

To address the contradiction between the dynamic changes of interspectral correlations in HSIs and the locality of spatial features, we design a dual-path collaborative attention mechanism. BFIN is shown in Fig. 3. Let the CNN feature of the source domain extracted by the feature extractor be x_c^s, the Transformer feature of the source domain be x_f^s, and the total feature of the source domain be x_{cf}^s. The target domains are x_c^t,

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x_f^t, and x_{cf}^t. In BFIN, the cross-domain information interaction between the source domain and the target domain is carried out first, which is called SSCA. From any x_{cf}^s and x_{cf}^t, the features are, respectively, decomposed into spectral and spatial subspaces to avoid the spectral–spatial coupling problem of traditional convolution. The formula for decomposition is defined as follows:

$$X_{spec} = reshape\ (X, (B, N, M)), \quad M = D/N. \tag{1}$$

Here, B represents the batch size, N represents the spectral dimension, M represents the spatial dimension, D represents the feature dimension, and X represents the total feature of the source domain or target domain that is input. The selection of the spectral subspace dimension N is based on the spectral characteristic analysis of the dataset. For hyperspectral data with L bands, we first calculate the correlation coefficient matrix $R \in \mathbb{R}^{L\times L}$ between bands and group the bands with high correlation through spectral clustering. Let the number of groups be G, then the initial $N_0 = G$. During the training process, we adjust it to a trainable hyperparameter and determine the optimal value through experiments. The spatial dimension M is obtained by dividing the total feature dimension D by N, that is, $M = D/N$, ensuring that the total information volume remains unchanged after reshaping. Then, the scaled dot product attention mechanism is adopted to operate on the reshaped source- and target-domain features, and the learnable temperature parameter τ is introduced for adjustment. The formula for this step is defined as follows:

$$\begin{aligned} F_{spec}(X) &= \text{softmax}\left(\frac{QK^T}{\sqrt{M}\tau}\right)V \\ \tau &= \text{sigmoid}(\omega_\tau). \end{aligned} \tag{2}$$

Here, M represents the spatial dimension, Q is the feature matrix of the source domain or the query matrix of the target domain, K and V are the key and value matrices of the target domain or the source domain, respectively, and $\sqrt{M}$ is the scaling factor. We defined a learnable temperature parameter τ so that the model can automatically learn the optimal range of temperature values, enabling the model to automatically adjust the intensity of attention focus at different training stages. It is worth noting that here we have designed a sparse attention constraint L_{sparse} to optimize the scaled dot product attention mechanism

$$L_{sparse} = \lambda \cdot \sum_{h=1}^{H} \|A_h\|_1. \tag{3}$$

Here, λ represents the sparse coefficient, H is the number of attention heads, and A_h is the attention weight matrix. It prompts the attention matrix to generate sparse patterns and automatically focus on key cross-domain feature pairs. Then, spatial attention is utilized to perform depth-separable convolution on the features of the source domain or the target domain to enhance locality, which is manifested as

$$F_{spa} = GELU\left(BN\left(Conv3\times 3_{depth}\left(X_{spec}\right)\right)\right). \tag{4}$$

$F(spa)$ is the spatial attention output. We conducted a preliminary gated fusion of the outputs of spectral attention and spatial attention. First, calculate the gating weight G through spectral focus $F_{spec}(X)$

$$G = \sigma\left(W_g avgpool\left(F_{spec}(X)\right)\right). \tag{5}$$

Here, $G \in [0,1]^N$, $W_g \in R^{N\times N}$, and N represents the spectral dimension. We conducted a preliminary gated fusion on the outputs of spectral attention and spatial attention to generate fusion features. The formula for integration is defined as

$$F_{fusion} = G \odot F_{spec} + (1-G)\odot F_{spa}. \tag{6}$$

Then, we adopted multilayer perceptron (MLP) with an expansion ratio of r for cross-domain feature fusion and ensured training stability through residual connection LayerNorm. The detailed formula is defined as follows:

$$\begin{aligned} F_{cross} &= MLP\left(\left[X \,\|\, F_{fusion}\right]\right) \\ Y &= LNorm\left(W_r\left[F_{cross}\,\|G\right] + X\right). \end{aligned} \tag{7}$$

Here, MLP is $R^{2D} \longrightarrow R^{rD} \longrightarrow R^D$. The above formula can clearly establish the attention mechanism in the spectral space, and then adaptively adjust the temperature coefficient τ and the gate G. Get the source-domain to target-domain characteristics of cross-domain attention $x_{cf}^{s\prime}$ to the source- and target-domain characteristics of cross-domain attention $x_{cf}^{t\prime}$. Then, the weights α_1 controlling the CNN features and α_2 controlling the Transformer features are obtained, respectively, by calculating the CNN gate and the Trans gate.

Bidirectional Dynamic Feature Preservation: To address the feature distribution differences caused by the fusion of multimodal features and cross-domain features, as well as the inability of traditional fixed-proportion fusion to adapt to sample specificity, we have designed a bidirectional dynamic feature preservation mechanism. For any input source- or target-domain feature matrix $Z \in R^{n\times d}$ and learnable feature transformation matrix $P \in R^{m\times d}$, the Wasserstein distance is calculated through the Sinkhorn algorithm [42], [43]. The core idea can be summarized as follows: by calculating the Wasserstein distance between the feature and the transformed feature space, the matching degree between the original feature and the target feature distribution is quantified. The smaller the distance, the closer the feature is to the ideal feature distribution after transformation, and the stronger its discriminability and the higher its importance. Define the cost matrix C_{ij}

$$C_{ij} = \left\|x_i - p_j\right\|_2 (1 \le i \le n, 1 \le j \le m). \tag{8}$$

Here, $|\cdot|_2$ is the Euclidean distance, which reflects the feature of the source domain (target domain) x_i as the source-domain (target-domain) target matrix p_j between matching the price. Then, the entropy regularized optimal transport is adopted, and the Sinkhorn algorithm is introduced to solve the following optimization problem:

$$W_\epsilon(Z,P) = \min_{T\in\Pi(n,m)} \langle T, C\rangle_F - \epsilon H(T). \tag{9}$$

Here, $\Pi(n,m)$ represents the feasible region of the transmission scheme, satisfying the marginal constraints: $T \in \mathbb{R}_+^{n\times m} \mid T\mathbf{1}_m = 1/n\mathbf{1}_n, T^\top\mathbf{1}_n = 1/m\mathbf{1}_m$. n and m follow a uniform distribution. $\epsilon > 0$ controls the regularization strength. A larger ϵ makes the transport matrix more uniform, while a smaller ϵ makes

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it sparser. $H(T) = -\sum_{i,j} T_{ij} \log T_{ij}$ is the entropy term. The above can be transformed into the actual calculation using logarithmic domain iteration, and finally, the transmission plan T and the Wasserstein distance $W = \langle T, C \rangle$ are obtained. Finally, the final importance is obtained through exponential mapping and normalization

$$I = \frac{e^{-w}}{max\left(e^{-w}\right) + 10^{-6}}. \tag{10}$$

Here, we add a small numerical value (10^{-6}) to the denominator to ensure the stability of the value. The importance range is (0, 1). The larger the value of I, the more important the feature is.

Multiscale consistency verification indicates that excellent features should maintain stable performance at different observation scales [44]. The feature robustness is verified through multiscale smoothing to filter out the abnormal responses caused by high-frequency noise [45]. The specific implementation method first performs multiscale average pooling (scale set) on the source-domain (target-domain) feature $f \in \mathbb{R}^d$, and then calculates the L_1 difference between each scale feature and the original feature

$$\begin{aligned} f_s &= \text{AvgPool1d}\left(f, kernel_size = s \right. \\ &\qquad \left., padding = s//2\right) \\ C(f) &= \frac{1}{|S|} \sum_{s \in S} \left(1 - \frac{\|f_s - f\|_1}{d}\right). \end{aligned} \tag{11}$$

Here, S is the multiscale pooling kernel, and scale 1 is equivalent to the original feature. We normalize $\|\cdot\|_1/d$ so that the result is within the range of [0, 1]. The larger the value of $C(f)$, the better the cross-scale consistency of the features is.

We effectively fuse the gating weights α_1 and α_2 generated by SSCA with the features screened by the bidirectional dynamic feature retention mechanism to generate an adaptive mask for the final feature screening. Take the source domain as an example, and the adaptive mask formula is defined as

$$\begin{aligned} Mask_{cnn} &= \left(\sigma\left(e^{k_{cnn}} \cdot (\alpha_1 - \gamma_{cnn}) \right.\right. \\ &\qquad \left.\left. + 2\left(\eta_{cnn} - 0.5\right)\right), 0, 1\right) \end{aligned} \tag{12}$$

$$\begin{aligned} Mask_{trans} &= \left(\sigma\left(e^{k_{trans}} \cdot (\alpha_2 - \gamma_{trans}) \right.\right. \\ &\qquad \left.\left. uad + 2\left(\eta_{trans} - 0.5\right)\right), 0, 1\right). \end{aligned} \tag{13}$$

Here, e^k is the slope coefficient, and its positivity is guaranteed through exponential transformation to control the transition steepness of the gate threshold interval. γ is the learnable threshold, which serves as the baseline for determining whether a feature is valid. η_{cnn} is the retention rate target. $\alpha_1 - \gamma_{cnn}$ determine whether the current feature exceeds the basic validity standard; $2(\eta_{cnn} - 0.5)$ maps the target retention rate to the interval $[-1, 1]$ as the bias term; the σ function implements soft judgment. The same applies to the target domain. In addition, we used the mean square error loss to calculate the retention rate loss for each feature and its generated adaptive mask, thereby optimizing the model. The formula is defined as

$$\begin{aligned} L_{mse} = \delta \left[\left(\frac{1}{B} \sum_{i=1}^{B} MAS\, K_{cnn}^i - \eta_{cnn} \right)^2 \right. \\ \left. + \left(\frac{1}{B} \sum_{i=1}^{B} MAS\, K_{trans}^i - \eta_{trans} \right)^2 \right]. \end{aligned} \tag{14}$$

Here, δ represents the loss weight and η represents the retention rate. The continuous optimization from hard constraints (target retention rate) to soft constraints (actual retention rate) is achieved through mse loss.

Mechanism Synergy Analysis: The SSCA mechanism explicitly models cross-domain interaction through spectral–spatial cross-attention, but its attention weights may be affected by noise and generate false associations. The Wasserstein distance measures feature differences at the global distribution level and provides robust alignment signals through the optimal transport theory, but it may ignore local structural consistency. Multiscale consistency verification focuses on the stability of local features and suppresses high-frequency noise by verifying the consistency of feature responses at different scales. Therefore, in the BFIN module, the collaborative design of SSCA, Wasserstein distance, and multiscale consistency verification has a clear motivation. These three elements achieve dynamic integration through a gating mechanism.

B. Dynamic Adaptive Topological Alignment

The purpose of dynamic adaptive topological alignment is to enable the model to adaptively align the structural features of the source domain and the target domain based on similarity differences during the training process. First, define the target-domain memory library: the target domain $M_t \in R^{M \times d}$, where M is the capacity of the memory library and d is the feature dimension. Let the feature space of the source domain be X_s, and the feature space of the target domain be X_t. Given a batch size for the characteristics of b $\{(f_s^i, f_t^i)\}_{i=1}^B$, we define the normalized features

$$M_t = \left\{\bar{m}_t^j\right\}_{j=1}^{M} \tag{15}$$

$$\bar{f}_s^i = \frac{f_s^i}{\left\|f_s^i\right\|_2}, \quad \bar{f}_t^i = \frac{f_t^i}{\left\|f_t^i\right\|_2}. \tag{16}$$

Here, f_s^i and f_t^i, respectively, are the characteristics of the source domain and target domain $\|\|_2$ for L2-norm, $\bar{f}_s^i$ and $\bar{f}_t^i$ are the normalized unit vectors, M is the memory capacity, and $\bar{m}_t^j$ is the history of memory normalized features. Feature normalization is mainly aimed at eliminating the differences in feature amplitudes, ensuring that similarity calculations only rely on direction, and guaranteeing that the subsequent cosine similarity is within the range of $[-1, 1]$. This kind of memory can preserve the long-term feature distribution between batches and reduce the gradient fluctuations caused by small-batch training. During the training alignment process, an adaptive parameter strategy is adopted to define the dynamic neighborhood size, and its formula is defined as

$$k(e) = min\left(k_{base} + \lfloor e/\tau \rfloor, k_{max}\right). \tag{17}$$

Here, the k_{base} as the initial number of neighborhoods, k_{max} for the maximum number of domains and prevent the smooth, τ to control the temperature coefficient of epoch growth range,

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e training for the current epoch. For every additional e units of τ, a neighborhood is added. During the early training, k_{base} ensures the correctness of the local topology. In the later training, k_{max} captures the global structure. To adjust the sharpness of the similarity distribution during the topological alignment process, an adaptive temperature coefficient $T(k)$ was designed

$$T(k) = max(0.1, T_0(1 + ln(k + 1))). \tag{18}$$

Here, T_0 is the base temperature. The logarithmic function ensures that the growth rate of $T(k)$ decreases as k increases, preventing excessive dispersion of weights due to the linear increase of temperature with k. High temperature $T(k)$ makes the weight distribution smooth, and all neighboring points obtain a relatively balanced gradient. In the early stage of training, it encourages the exploration of more potential alignment relationships. Low temperature $T(k)$ can concentrate the weights, with only the most similar points dominating the gradient update; that is, the focus is on high-confidence alignment in the later stage of training.

The traditional contrast loss algorithm only uses the current sample to calculate the similarity, resulting in a large variance of gradient estimation and making it difficult to capture the global feature distribution [46], [47]. The abovementioned issues are reflected in the continuous alignment process of source-domain features and target-domain features, where there may be distribution offsets (such as differences in illumination and viewing angles) between the features of the source domain and the target domain. To improve this issue, we have established a memory update mechanism. Memory update can be regarded as a sliding window estimation of feature distribution, which is equivalent to establishing a dynamic prototype dictionary and providing a more stable distribution estimation. The update strategy adopts a momentum update and gradually integrates the potential structure shared across domains. The momentum update strategy is defined as

$$m_t^j \leftarrow \eta(b)\, \bar{f}_t^i + (1 - \eta(b))\, m_t^j. \tag{19}$$

Here, m_t^j is the j-th feature in the target-domain memory, used for the moving average of historical target-domain features; f_t^i is the ith original feature of the source domain in the current batch, $\bar{f}_t^i$ is the normalized feature of the source domain, and b indicates the batch. The momentum coefficient $\eta(b) = min(0.5, 0.3 + 0.01b)$ increases linearly with the training period. The initial momentum focuses on historical information in the early stage and on current features in the later growth stage. The contributions of both new and old features have been balanced. To capture the instantaneous feature relationship, reflect the alignment status of the current batch of the model, and introduce historical distribution information to provide global consistency constraints across batches, we have established an enhanced similarity matrix S

$$S = \left[S_{current}(i, j) \,\middle|\middle|\, S_{memory}\right] \in R^{B\times(B+M)} \tag{20}$$

$$S_{current}(i, j) = \left\langle \bar{f}_s^i, \bar{f}_t^j \right\rangle,\ S_{memory}(i, j) = \left\langle \bar{f}_s^i, \bar{m}_t^j \right\rangle. \tag{21}$$

Here, $\langle \cdot, \cdot \rangle$ cosine similarity (dot product) is used to calculate the directional consistency between feature vectors, B represents the batch size, and M represents the memory capacity. Each row of the matrix corresponds to the similarity between a source-domain feature and all target-domain features (current + historical). When the sample distribution within a batch is unbalanced, the memory can supplement the similarity calculation of the missing modes. Meanwhile, the concatenation operation $\|$ expands the similarity search space and enhances the homogeneity robustness. Specifically, for each anchor point feature $\bar{f}_s^i$, first perform temperature scaling $[\tilde{S}(i, :) = S(i, :)/T(k)]$, adaptively adjusting with the number of neighborhoods k. Then, perform top-k neighborhood selection $[N_k(i) = TopK(\tilde{S}(i, 1 : B), k)]$, select k most relevant target-domain feature indicators for each source-domain feature in the current batch, and perform weighted alignment loss. The formula for weighted alignment loss is defined as follows:

$$L_{align}^i = \sum_{j\in N_k(i)} \omega_{ij}(1 - S_{current}(i, j)) \tag{22}$$

$$\omega_{ij} = \frac{exp\left(\tilde{S}(i, j)\right)}{\sum_{l\in N_k(i)} exp\left(\tilde{S}(i, l)\right)}. \tag{23}$$

Here, w_{ij} is the weight normalized by softmax, reflecting the importance of the neighborhood points; and $S_{current}(i, j)$ represents the original similarity between the source-domain feature i and the target-domain feature j. High similarity match $S(i, j) \longrightarrow 1$ gains greater weight and dominates the gradient update. The $1 - S$ transforms the problem of maximizing similarity into minimizing loss. The direction of the gradient is jointly determined by the similarity gradients of the weighted neighborhood points. The temperature $T(k)$ controls the gradient amplitude by adjusting w_{ij}. Overall, the dynamic adaptive topological alignment loss realizes the construction of the similarity matrix of feature pairs, then the selection of neighborhood k, and ultimately the weighted loss. In addition, the coarse-to-fine alignment of the training process was achieved through $k(e)$ and $T(k)$. The final loss L_{tal} is the average loss of the batch

$$L_{tal} = \frac{1}{B}\sum_{i=1}^{B} L_{align}^i. \tag{24}$$

The features of the source domain and the target domain establish a similar topological structure from the source domain to the target domain in the feature space with memory characteristics. This method aims to achieve DA at the feature level and has stronger constraints. Dynamic adaptive topological alignment builds a topological structure that aligns the source domain with the target domain by dynamically adjusting the neighborhood k and temperature to adapt to the complex training process. Meanwhile, historical features are stored in the target-domain memory, updated with momentum, enhancing the continuity of the topological structure between the source domains and demonstrating strong domain adaptability.

C. Topology-Aware Multiview Consensus Pseudo-Label Optimization

To obtain multiperspective prediction information of the target domain, we performed data-enhanced random flipping

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operations on the data of the source domain and the target domain. Random flipping can horizontally or vertically flip samples to increase the diversity of the data. From the first perspective, the prediction of the original source domain and the target domain is *Pred_v*0; from the second perspective, the prediction of the enhanced data of the source domain and the target domain is *Pred_v*1; from the third perspective, the prediction of the enhanced data of the target domain and the source domain is *Pred_v*2. The overall flowchart of TA-MVCPO is shown in Fig. 1.

For the target-domain sample X^t, the classifier output of the three perspectives is defined as $\{f_v(x_t)\}_{v=0}^2$, and its prediction probability distribution is $\{p_v(y|(x_t))\}_{v=0}^2$. We adopted a pseudo-label screening method, which follows two hierarchical consensus principles.

1) *Strict Consensus Principle:* We first screen samples whose predictions from all three perspectives are consistent and whose confidence levels all exceed the dynamic threshold. We define the strict consensus principle as

$$M_s = \left\{ x_t \,\middle|\, \arg\max_y \, p_v\left(y|x_t\right) = y^*, \; \max \, p_v\left(y|x_t\right) > \tau\left(\eta, \Delta\right), \; \forall v \right\}. \tag{25}$$

Here, $\eta \in [0, 1]$ represents the training progress coefficient, x_t is the feature vector of the target-domain samples, y^* is the candidate pseudo-label category, p_v is the prediction probability distribution of perspective v, f_v is the classifier of the v-th perspective, and Δ is the topological difference score. The threshold $\tau(\eta, \Delta)$ is adaptively adjusted according to the training progress $\eta = (epoch)/(E_{max})$ and topological differences Δ

$$\tau\left(\eta, \Delta\right) = \tau_{max} - \left(\tau_{max} - \tau_{min}\right)\eta + \alpha\Delta. \tag{26}$$

Here, α represents the topological difference sensitivity coefficient to suppress the generation of pseudo-labels of topologically inconsistent categories. τ_{max} is the maximum initial threshold to ensure high confidence, and τ_{min} is the lower limit of the minimum threshold to prevent over-filtering. When the predictions from the three perspectives are consistent, and the confidence levels all exceed the dynamic threshold $\tau(\eta, \Delta)$, pseudo-labels with strict consensus are generated.

2) *Loose Consensus Principle (Dual-Perspective Verification):* Under the condition of dual-perspective consistency, we have enhanced the confidence threshold to prevent the problem of excessive filtering or ineffective screening of the strict consensus principle in a certain category. The formula of this principle is defined as

$$M_p = \bigcup_{i<j} \left\{ x_t \,\middle|\, \text{argmax} p_i = \text{argmax} p_j \wedge \frac{\max p_i + \max p_j}{2} > 1.2\tau\left(\eta\right) \right\}. \tag{27}$$

Here, $\bigcup_{i<j}$ represents the combination of traversing all perspectives. The final pseudo-label set is $\hat{y} = \{\hat{y}_t | x_t \in M_s \cup M_p\}$.

When there is a significant difference between the topological structure of the pseudo-label and that of the source domain, we designed a topological adaptive enhancement principle to dynamically enhance the strictness of the threshold. Therefore, topological feature extraction is performed on the feature set $H^c = \{h_i^c\}_{i=1}^n$ of category c of category c. The formula for calculating the persistence graph through the Vietoris–Rips [48], [49], [50] complex is defined as

$$PD_k^c = VR\left(H^c, \epsilon_{max}\right), \quad k \in \{0, 1\}. \tag{28}$$

Here, ϵ_{max} represents the threshold of the maximum connection distance, H^c is the feature set of category c, and PD_k^c is the persistent graph of dimension k. (b, d) is the birth and death coordinates of the topological feature. Then, the Wasserstein amplitude distance is calculated for the difference score quantification. The greater the difference, the more significantly the topological structure of the feature space of this category deviates from the source domain. The calculation formula for the difference score is defined as

$$\Delta^c = \frac{1}{|PD^c|} \sum_{(b,d)\in PD^c} |d - b| \cdot \left\| PD_s^c\left(b, d\right) - PD_t^c\left(b, d\right) \right\|_2. \tag{29}$$

Here, PD^c is the persistence graph of category c, representing point set $\{(b_i, d_i)\}$, where b_i is the birth time of the feature and d_i is the death time. $d - b$ represents the sustained lifespan of the feature. The longer the lifespan of the topological feature, the greater its contribution to the difference (more stable structure). PD_s^c is the persistent graph of the source-domain category c. $\|\cdot\|_2$ is the Euclidean distance, quantifying the coordinate difference of the topological features between the source domain and the target domain. $\|PD_s^c(b, d) - PD_t^c(b, d)\|_2$ represents the coordinate difference, which measures the spatial distribution offset of similar features in the source domain and the target domain. The global topological difference degree is $\bar{\Delta} = 1/C\sum_{c=1}^{C}\Delta^c$. It integrates the local differences of all categories, reflects the overall topological inconsistency between domains, and divides by the number of categories C to avoid the bias caused by category imbalance.

Threshold Adaptation: The threshold correction mechanism is implemented under the condition of obtaining topological differences. Its purpose is to increase the threshold when the topological differences are large, filter out more suspicious samples, and gradually relax the threshold in the later stage of training to utilize more data. Therefore, topology-aware dynamic adjustment terms are introduced for threshold correction. The formula η_f is defined as follows:

$$\eta_f = \left(\eta, \bar{\Delta}\right) = \tau\left(\eta\right) + \beta\bar{\Delta}. \tag{30}$$

Here, β balances the classification confidence and topological consistency, and $\tau(\eta)$ is the base threshold that decays with the training progress.

In the TA-MVCPO strategy, multiview confidence screening combines the topological differences between source-domain labels and target-domain pseudo-labels, enabling the screening of pseudo-labels with high confidence and fine-tuning of the model.

The complete training pipeline of the proposed TAGIN framework is formally described in Algorithm 1, which details:

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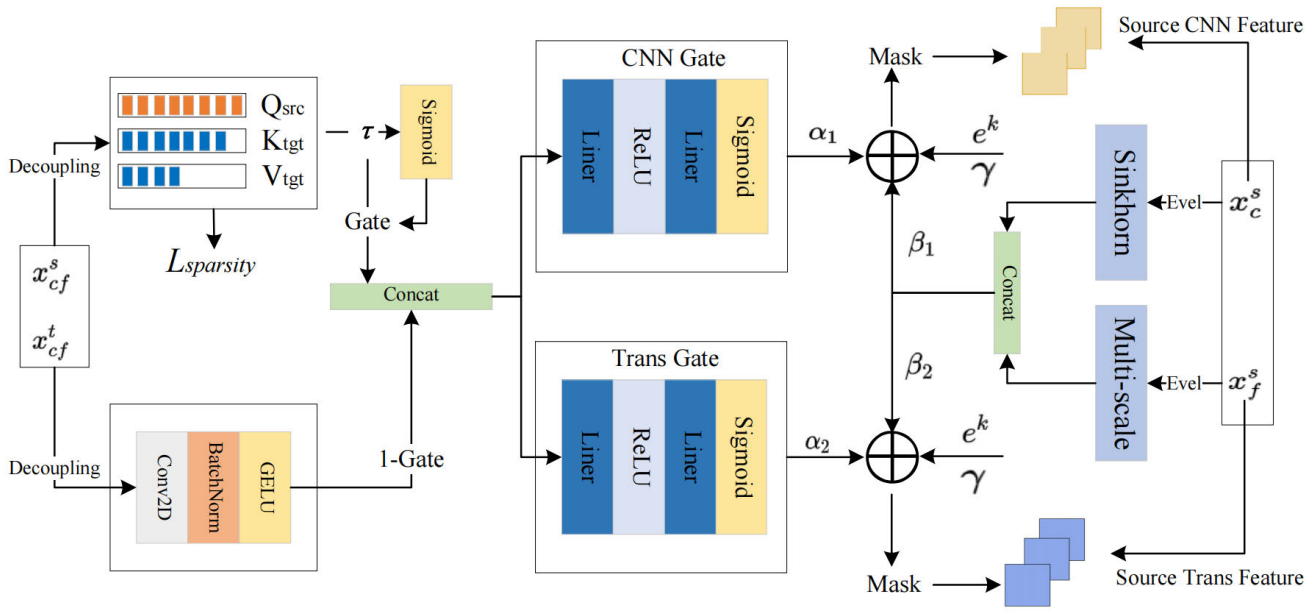


Fig. 3. Perspective of cross-domain feature interaction from the source domain to the target domain in BFIN.

Algorithm 1 Training Pipeline of TAGIN

Input: Source domain samples X^s and labels Y^s,target domain samples X^t, epochs

Output: Feature extractor E

for epoch in 1,.....epochs **do**
$\bar{X}^s = Augment(X^s)$
$\bar{X}^t = Augment(X^t)$
$(X^s, Y^s) \longrightarrow L^s_{cls}$
$(X^s, X^t) \longrightarrow L_{tal}$
$(X^s, X^t) \longrightarrow Pred_v0$
$(\bar{X}^s, X^t) \longrightarrow Pred_v1$
$(X^s, \bar{X}^t) \longrightarrow Pred_v2$
$MVCPO(Pred_vi, \tau(\eta))$through Eq.(25), (26) and (27)$\longrightarrow \hat{y}_init$
if epoch % $Verification_NUM == 0$ **then**
$\Delta \longleftarrow TA(X^s, Y^s, X^t, \hat{y}_init)$ through Eq. (28), (29)
$\hat{y}_final \longleftarrow MVCPO(Pred_vi, \tau(\eta, \Delta))$add Eq.(30)
Fine-tuning $\hat{y}_final$
end if
end for

1) the iterative pseudo-label refinement process through multiview consensus voting and 2) the topological consistency validation mechanism using persistent homology analysis.

IV. Experiment and Discussion

A. Data Description

1) Houston: Houston2013 dataset comprises 349 × 1905 pixels with a spatial resolution of 2.5 m. It features 144 spectral bands and includes 15 distinct land cover categories. Houston2018 dataset measures 209 × 955 pixels with a spatial resolution of 1 m, containing 48 spectral bands that encompass 20 categories. To maintain consistency across comparisons, 48 spectral bands and a region of 209 × 955 pixels were selected from Houston2013 dataset. Both datasets include seven overlapping categories. The list of classes and the number of samples are presented in Table I. Fig. 4 displays the false-color images and ground-truth maps.

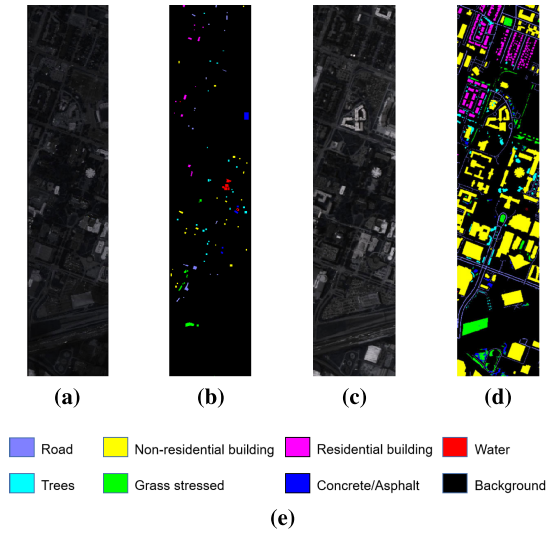


Fig. 4. Houston image. (a) False-color image of Houston in 2013. (b) Ground reality map of Houston in 2013. (c) False-color images of Houston in 2018. (d) 2018 Houston physical map. (e) Legend.

TABLE I
Number of Samples in Houston

Houston			
No.	Class	2013(Source)	2018(Target)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
	Total	2530	53200

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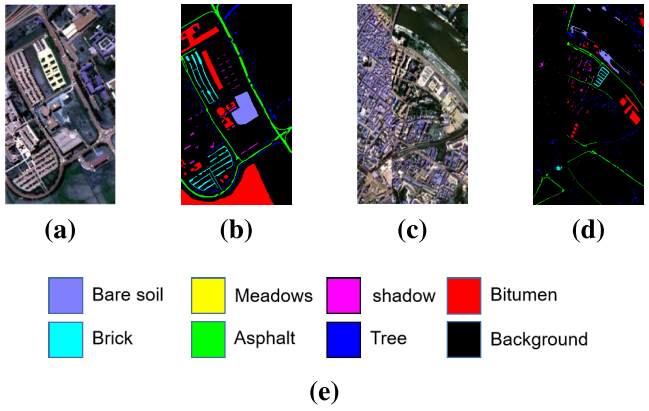


Fig. 5. Pavia image. (a) False-color image of UP. (b) Ground-truth map of University of Pavia (UP). (c) False-color image of Pavia Centre (PC). (d) Ground-truth map of PC. (e) Legend.

TABLE II
NUMBER OF SAMPLES IN PAVIA

Pavia			
No.	Class	PaviaU(Source)	PaviaC(Target)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadows	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

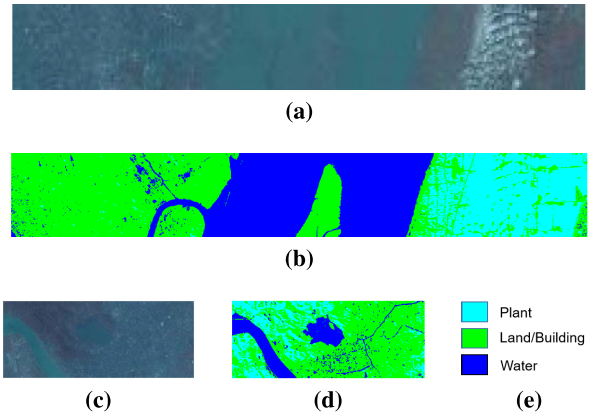


Fig. 6. SH2HZ image. (a) False-color image of Shanghai. (b) Ground-truth map of Shanghai. (c) False-color image of Hangzhou. (d) Ground-truth map of Hangzhou. (e) Legend.

2) Pavia: The datasets from Pavia were acquired using the Reflective Optics System Imaging Spectrometer (ROSIS). The University of Pavia dataset includes 610 × 610 pixels and 103 spectral bands, whereas Pavia Centre dataset contains 1096 × 1096 pixels and 102 spectral bands. To standardize data handling, the former was resized to 610 × 315 pixels, and the latter to 1906 × 715 pixels. Additionally, the final spectral band was removed from the University of Pavia dataset to ensure uniformity. The list of classes and the number of samples are presented in Table II. Fig. 5 displays the false-color images and ground-truth maps.

3) SH2HZ: Shanghai and Hangzhou datasets were collected using the EO-1 Hyperion hyperspectral sensor. After excluding bands with poor quality, 198 bands were retained for each dataset. Shanghai dataset is formatted as 1600 × 230 pixels, and Hangzhou dataset as 590 × 230 pixels. Both datasets feature 198 bands. The list of classes and the number of samples are presented in Table III. Fig. 6 displays the false-color images and ground-truth maps.

TABLE III
NUMBER OF SAMPLES IN THE SH2HZ

SH2HZ			
No.	Class	Shanghai(Source)	Hangzhou(Target)
1	Water	123123	18043
2	Land/building	161689	77450
3	Plant	83188	40207
Total		368000	135700

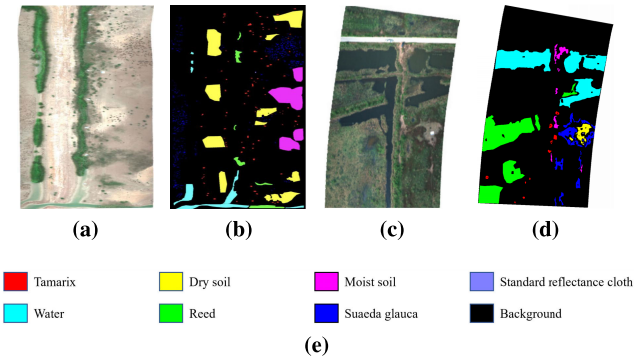


Fig. 7. NC image. (a) False-color image of NC12. (b) Ground-truth map of NC12. (c) False-color image of NC13. (d) Ground-truth map of NC13. (e) Legend.

TABLE IV
NUMBER OF SAMPLES IN THE NC

NC			
No.	Class	NC12(Source)	NC13(Target)
1	Suaeda glauca	14117	14225
2	Reed	15603	56922
3	Water	36291	63007
4	Tamarix	9257	1954
5	Moist soil	68497	4910
6	Dry soil	120073	4427
7	Standard reflectance cloth	75	49
Total		263913	145494

4) NC: The NC12 and NC13 datasets [51] were collected using a 12-mm focal length top-wall nanohyperspectral imaging sensor mounted on a DJI M600 drone platform within Shandong Yellow River Delta National Nature Reserve. This sensor recorded 270 spectral bands spanning wavelengths from 400 to 1000 nm. The NC12 dataset was acquired between 12:27 and 13:10 on September 23, 2020, under clear, cloudless weather conditions. The spatial resolution of the NC12 dataset is approximately 0.182 m, as the drone flew at an altitude of 300 m. Additionally, the NC13 dataset was acquired between 14:47 and 15:20 on September 24, 2020, under overcast conditions. With the uncrewed aerial vehicle (UAV) flying at 300-m altitude, the spatial resolution of the NC13 dataset is also approximately 0.182 m. These scenes encompass seven consistent land cover categories. Table IV displays the class names and sample counts in both the source and target domains for the NC12 →NC13 task. Fig. 7 presents the false-color images and corresponding ground-truth maps.

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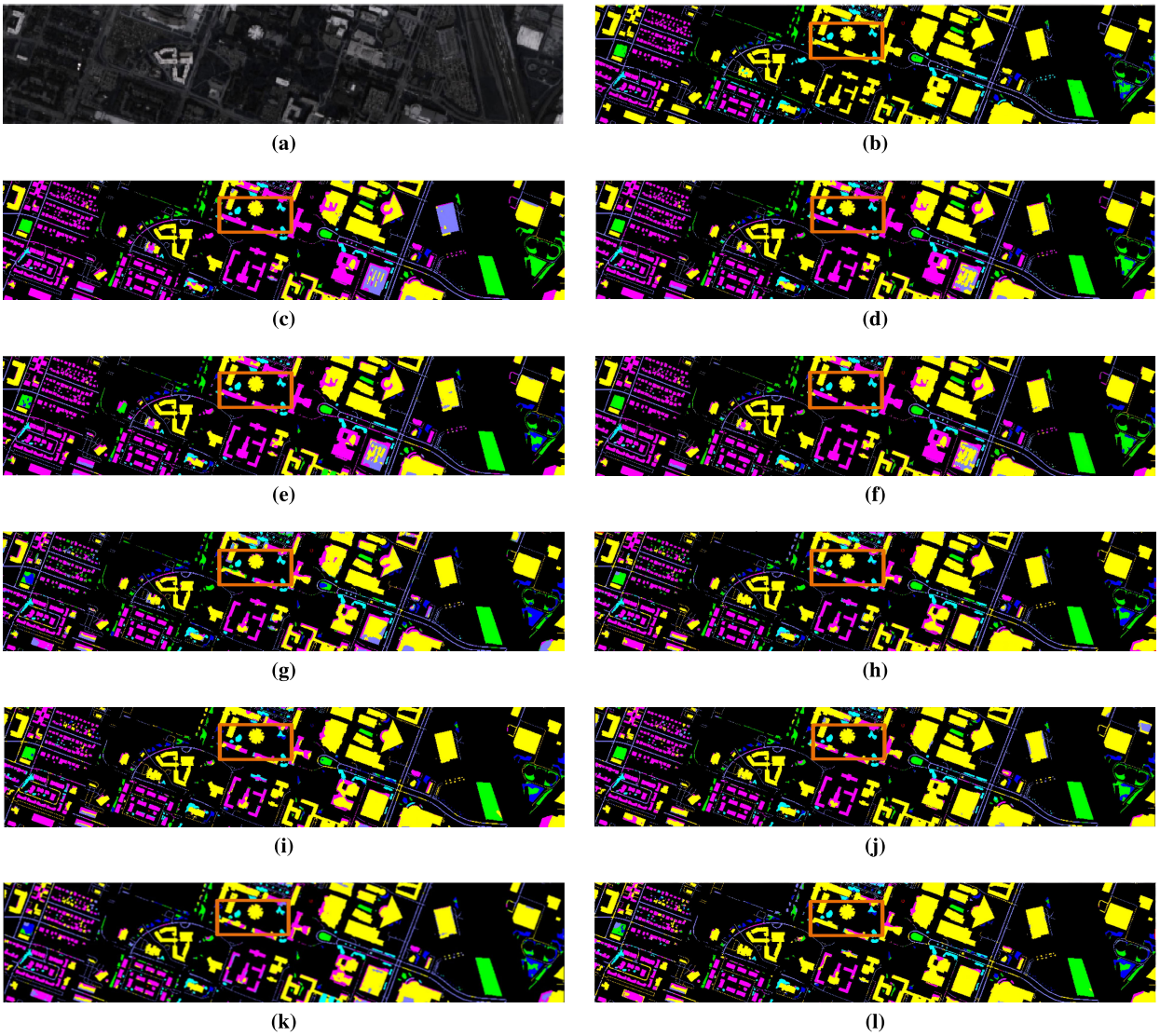


Fig. 8. Houston. (a) False-color image. (b) Ground truth. (c) DANN. (d) EDDMM. (e) CDA. (f) CLDA. (g) TAADA. (h) SCLUDA. (i) MLUDA. (j) MSDA. (k) ASAL. (l) Ours.

B. Experimental Setup

The experiments were implemented using the PyTorch framework to operate the TAGIN. A total of 100 epochs were set for the training phase. The mini-batch SGD optimizer with a momentum of 0.9 was used for backpropagation. The initial learning rate lr0 was established at 0.1 for tasks involving Houston dataset, applying a learning rate decay strategy defined by $l_{r\theta} = l_{r0}((1 + a\theta))^b$, where $a = 10$, $b = 0.75$, and θ represents a linear progression from 0 to 1 during training. For Pavia, SH2HZ, and NC datasets, the learning rates are set to 0.004, 0.001, and 0.01, respectively. The patch sizes selected for these tasks are 7×7, 11×11, 1×1, and 1×1, respectively, and the batch size of the tasks is consistently 32. The weight decay parameter was set at $5e^{-4}$ for all tasks. In the experiment, we selected 180 samples for each category of the source domain and trained with all samples from the target domain.

To evaluate the effectiveness of this method, the experimental results for Houston, Pavia, SH2HZ, and NC datasets were compared with those obtained using techniques such as DANN [17], deep-metric learning-based feature embedding (EDDMM) [52], classwise distribution adaptation (CDA) [53], confident learning-based DA (CLDA) [54], TAADA [55], supervised contrastive learning-based unsupervised DA (SCLUDA) [18], multilevel unsupervised DA (MLUDA) [21], masked self-distillation DA (MSDA) [56], and adaptive smooth adversarial learning (ASAL) [57].

C. Experimental Results

1) Houston: Table V presents the performance metrics of various methods on Houston dataset. The performance indicators of DANN are relatively low, especially the Kappa coefficient, and there are problems with random stability. The three methods of EDDMM, CDA, and CLDA have gradually improved the disadvantages of the first two methods. However, from an overall perspective, the Kappa coefficient still has not exceeded 60%, and its credibility is questionable. The classification accuracy of the first five methods varies

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TABLE V

CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON HOUSTON

Class	DANN	EDDMM	CDA	CLDA	TAADA	SCLUDA	MLUDA	MSDA	ASAL	Ours
1	74.13	68.14	**95.08**	63.03	88.99	80.03	83.26	71.69	79.44	73.02
2	74.69	**85.45**	51.96	84.85	48.69	69.75	75.69	67.01	82.92	80.87
3	68.62	57.86	62.52	**72.18**	60.38	64.13	54.13	64.34	61.00	62.87
4	96.59	79.55	95.45	94.55	93.18	92.61	94.55	77.64	**96.97**	93.94
5	76.64	**97.77**	70.35	93.59	77.84	89.38	84.28	89.72	91.57	89.26
6	57.76	56.62	82.74	62.16	76.68	73.56	79.33	81.56	73.18	**85.53**
7	49.51	75.97	26.29	**79.26**	77.95	57.35	65.74	73.90	77.78	60.41
OA	61.23±4.65	66.09±1.83	71.18±1.31	70.02±2.94	73.85±3.57	75.50±1.77	76.64±1.18	78.96±2.27	76.03±1.99	**79.96±1.51**
AA	71.14±3.50	74.48±1.66	69.20±3.16	78.53±2.06	74.82±2.64	74.32±4.93	76.71±1.99	76.31±1.77	71.39±1.49	**77.56±1.97**
$\kappa \times 100$	45.21±4.60	53.10±2.18	51.85±3.23	57.44±3.11	60.39±4.25	61.00±3.38	63.62±1.47	66.52±1.93	59.09±2.43	**67.15±1.82**

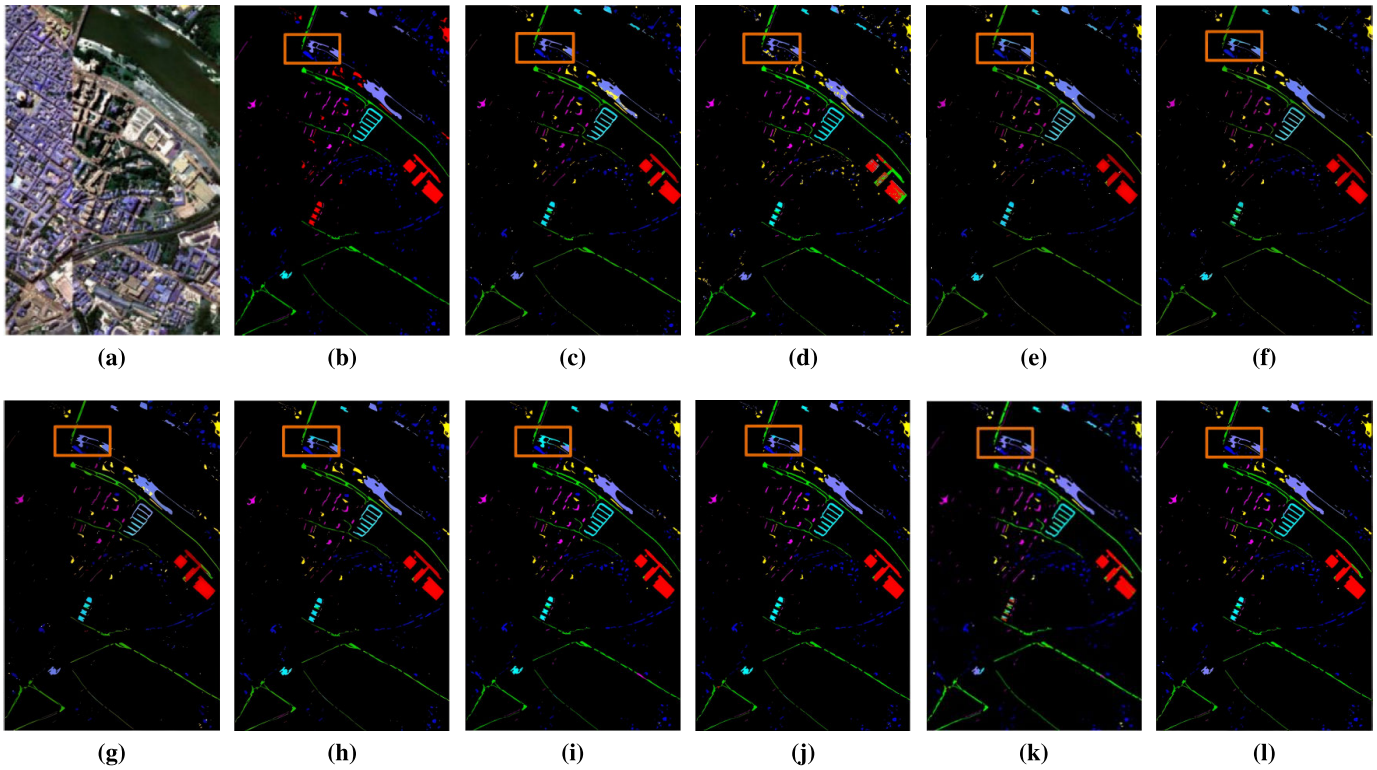


Fig. 9. Pavia. (a) False-color image. (b) Ground truth. (c) DANN. (d) EDDMM. (e) CDA. (f) CLDA. (g) TAADA. (h) SCLUDA. (i) MLUDA. (j) MSDA. (k) ASAL. (l) Ours.

TABLE VI

CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON PAVIA

Class	DANN	EDDMM	CDA	CLDA	TAADA	SCLUDA	MLUDA	MSDA	ASAL	Ours
1	75.09	**98.21**	92.02	96.58	88.78	95.97	93.02	93.44	94.22	96.18
2	78.23	96.63	77.63	**99.68**	98.62	98.44	96.36	98.19	87.61	97.46
3	80.69	72.40	86.90	78.07	65.74	**97.03**	96.81	79.92	96.42	71.89
4	73.08	67.94	78.17	80.07	72.05	85.70	85.42	83.87	**85.54**	82.29
5	98.28	**100**	99.94	99.99	99.90	98.42	99.91	99.84	99.24	98.20
6	86.69	25.46	61.81	86.96	**94.65**	94.33	94.45	86.47	85.09	94.23
7	73.51	83.92	71.60	90.59	96.30	78.13	79.97	96.95	80.33	**98.60**
OA	78.16±4.08	82.50±0.50	80.51±1.02	90.48±2.26	88.95±2.82	91.91±1.44	91.26±0.53	92.14±0.90	91.34±5.59	**92.50±0.36**
AA	80.79±2.52	77.80±0.56	81.15±1.81	90.28±1.96	88.01±3.31	**91.93±0.90**	91.15±0.80	91.23±1.73	89.78±0.97	91.29±0.56
$\kappa \times 100$	74.16±4.62	78.73±0.54	76.68±1.22	89.71±2.72	86.76±3.33	90.93±0.51	89.63±0.63	90.93±1.22	86.27±1.50	**91.04±0.43**

greatly in each category, which is highly related to the imbalance among the categories. The emergence of TAADA and SCLUDA has, to some extent, alleviated the problems existing in the abovementioned several methods. The classification differences of TAADA among various categories are gradually decreasing. The overall accuracy rate of SCLUDA has reached 75%, which is related to its adoption of a high-confidence fine-tuning strategy in the target domain. MLUDA gradually improved the Kappa coefficient on the basis of the former through the cross-attention mechanism, with the index reaching 63%. MSDA employs distillation masking technology to enhance DA capabilities, and its overall accuracy

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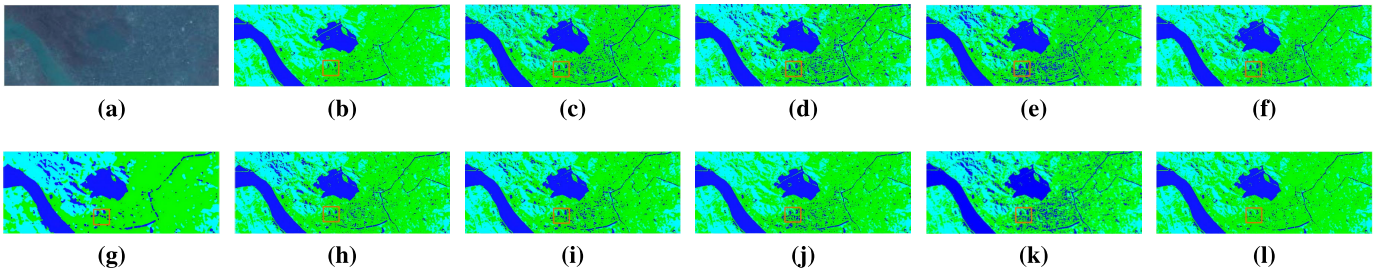


Fig. 10. SH2HZ. (a) False-color image. (b) Ground truth. (c) DANN. (d) EDDMM. (e) CDA. (f) CLDA. (g) TAADA. (h) SCLUDA. (i) MLUDA. (j) MSDA. (k) ASAL. (l) Ours.

TABLE VII
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE SH2HZ

Class	DANN	EDDMM	CDA	CLDA	TAADA	SCLUDA	MLUDA	MSDA	ASAL	Ours
1	86.20	94.70	97.95	96.60	94.80	94.21	94.39	96.22	96.69	**98.87**
2	86.46	88.80	83.57	89.41	77.43	84.63	85.21	95.00	88.18	**95.04**
3	79.51	81.48	89.28	88.78	82.75	**99.45**	98.55	86.69	83.06	87.76
OA	84.37±2.32	87.42±6.30	87.18±0.84	90.18±0.98	81.31±1.70	91.19±0.87	91.76±1.88	92.17±1.73	87.80±1.10	**93.38±1.16**
AA	84.06±6.36	88.33±5.43	90.27±0.77	91.60±1.35	84.99±0.69	92.76±0.74	92.32±1.67	91.96±1.52	89.31±0.05	**93.89±1.09**
$\kappa \times 100$	73.06±4.20	78.12±6.91	78.39±1.35	83.05±1.56	68.73±2.20	86.56±1.31	84.44±2.80	87.13±2.61	78.89±1.60	**88.41±2.02**

rate has increased to 78%. ASAL has significantly improved the classification accuracy of residential and nonresidential buildings in Houston data through smooth adversarial training, label smoothing, and the adoption of a two-stage training strategy. Our method has also made corresponding improvements in the indicators of the previous several methods. For instance, through feature-level alignment of topological structure, the effect of DA has been enhanced, especially the indicators of overall accuracy (OA), and the Kappa coefficient can better reflect this. Fig. 8 shows the visual classification graphs generated by different algorithms. Among other classification algorithms, nonresidential buildings are mostly wrongly classified as residential buildings. Only our method classifies them with relatively accurate results. Because these two categories are relatively similar in spatial structure or spectral characteristics, incorrect classification is quite common.

2) Pavia: Table VI presents the performance metrics of various methods on Pavia dataset. The three methods, DANN, EDDMM, and CDA, each have their own advantages in terms of OA, average accuracy (AA), and Kappa coefficient. The CLDA method broke through 90% for the first time in the OA and AA indicators, achieving good results. The three methods, TAADA, SCLUDA, and MLUDA, have achieved outstanding results both in individual category indicators and in OA, AA, and Kappa coefficients. Based on the previous method, MSDA has improved the distinguishability of the target-domain features through distillation masks. This innovation has led to further improvements in three indicators. ASAL introduces multiscale covariance analysis to capture the category distribution relationships at different scales. It performs exceptionally well in distinguishing spectral details, for instance, achieving a relatively high classification accuracy for bricks and asphalt in Pavia dataset. Our method remains highly effective in implementing cross-domain feature-level alignment, and the performance of the three metrics remains optimistic. Fig. 9 shows the visual classification graphs generated by different algorithms. Since Brick and Bare soil are extremely similar in essence, the spatial and spectral characteristics they reflect are difficult to distinguish. Other mainstream algorithms do not classify them completely correctly, while our method classifies them successfully and correctly. The role of our method in constructing subtle topological differences and analyzing spectral semantic information should not be underestimated, thus demonstrating significant importance in difficult samples that are hard to classify.

3) SH2HZ: Table VII shows the results of the experiment on the SH2HZ dataset. Compared with the previous two datasets, the performance of each aspect on this dataset is generally better. Under the condition of supervised contrastive learning, SCLUDA further distinguishes the differences between classes and reduces the differences within classes. Its OA and AA indicators have exceeded 90% after fine-tuning with high-confidence pseudo-labels, but the Kappa coefficient remains relatively low. MLUDA has achieved relatively thorough alignment at the input level through image transfer technology and the application of cross-domain attention mechanisms for features, but there is still considerable room for performance improvement. The entropy-weighted minimum category confusion loss proposed by ASAL adjusts the confidence level of the model output through dynamic temperature scaling. It enables the model to pay more attention to highly uncertain (high-entropy) samples and enhances its learning ability for difficult samples. Compared with these methods, other methods still have their own merits, but the overall performance still needs to be improved. In contrast, our approach delves deeply into the potential of topological structures in DA alignment, achieving feature-level DA, which is reflected in a significant improvement in the Kappa coefficient. Therefore, it can be concluded overall that TAGIN has strong adaptability on this dataset. Fig. 10 shows the visual classification graphs generated by different algorithms. The regions where Water

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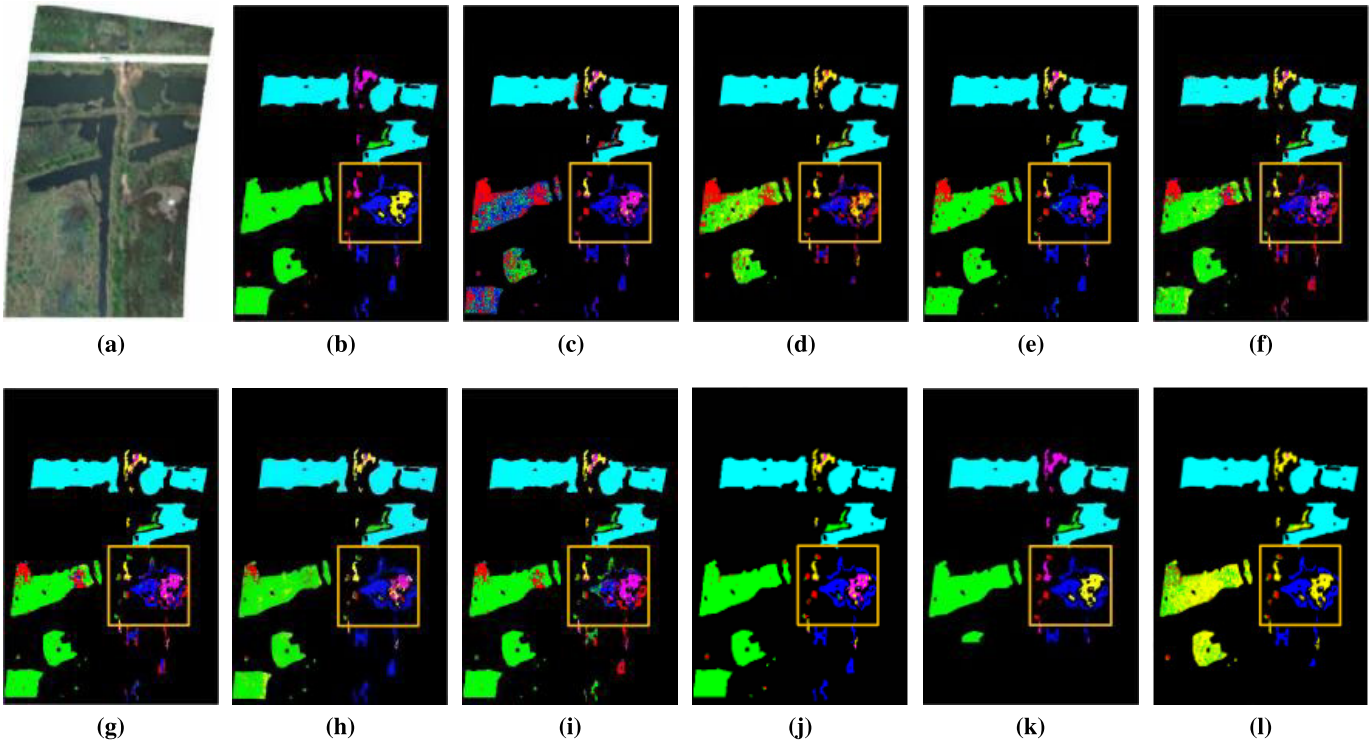


Fig. 11. NC. (a) False-color image. (b) Ground truth. (c) DANN. (d) EDDMM. (e) CDA. (f) CLDA. (g) TAADA. (h) SCLUDA. (i) MLUDA. (j) MSDA. (k) ASAL. (l) Ours.

TABLE VIII
CLASSIFICATION RESULTS (VALUES ± STANDARD DEVIATION) ON THE NC

Class	DANN	EDDMM	CDA	CLDA	TAADA	SCLUDA	MLUDA	MSDA	ASAL	Ours
1	37.77	68.40	52.44	58.39	92.82	80.28	33.15	55.08	96.31	**98.97**
2	77.13	46.96	72.58	72.01	79.49	87.06	91.04	94.97	61.93	**97.00**
3	20.20	70.35	98.01	97.99	99.96	93.79	99.90	**99.98**	99.96	99.78
4	56.55	31.02	45.29	55.55	67.22	35.60	31.53	45.79	**70.50**	40.11
5	31.53	2.75	19.61	35.23	30.22	11.41	18.29	17.10	**54.13**	34.04
6	7.99	25.20	11.02	0.4	1.25	16.26	1.15	14.35	0.00	**28.73**
7	68.37	75.71	**100**	98.99	80	97.84	**100**	**100**	**100**	**100**
OA	44.71±18.80	56.83±25.39	77.60±4.45	80.13±6.08	85.96±2.58	85.00±4.17	83.56±2.45	87.50±4.55	86.75±1.96	**93.42±0.54**
AA	42.79±10.09	45.77±10.43	56.99±9.60	60.37±10.63	65.57±5.18	65.54±5.38	55.01±2.54	61.04±6.09	68.98±0.74	**71.23±3.91**
$\kappa \times 100$	29.43±22.74	43.78±26.09	67.56±6.10	71.37±8.27	79.28±3.43	77.76±5.70	74.82±3.52	80.92±6.65	71.35±7.72	**89.82±0.83**

and Land are located are rather complex, which leads to a relatively high degree of discrimination. Other algorithms have not effectively distinguished these two categories but have mixed in incorrect classifications. In contrast, our method has a higher degree of discrimination compared to the real image.

4) NC: Table VIII shows the results of the experiment on the NC dataset. NC12 data and NC13 data are a type of cross-climate (weather) and cross-scenario classification data. Under this condition, DANN, EDDMM, CDA, and CLDA did not achieve ideal classification results. Their classification accuracy in categories 1, 5, and 6 is so poor that their performance in OA, AA, and Kappa indicators is uneven. TAADA, SCLUDA, and MLUDA have seen significant improvements. MSDA proposes a thinking space-spectral masking strategy to mask HSI at the fragment level. It retains the central pixel to avoid masking and damaging the category information. This advantage is demonstrated in the context of cross-climate data. The ASAL method introduces deformable convolution, enabling the convolution energy to adaptively adjust the sampling position and better capture irregular targets. This performance in the NC data results is similar to that of MSDA. Our TAGIN method has seen a significant improvement compared with others, especially in terms of OA and Kappa coefficients. This indicates that the gating mechanism based on the BFIN we proposed is more thorough in deep feature extraction under cross-climate conditions. Furthermore, the dynamic adaptive topological alignment loss we proposed can effectively solve the DA problem in the data feature alignment under this condition. Fig. 11 shows the visual classification graphs generated by different algorithms. In the visual classification of Suaeda glauca, Moist soil, and Dry soil in the figure, our method is closer to the real classification effect. In particular, Moist soil and Dry soil are produced in different climates, respectively, and other methods have produced incorrect classifications in Moist soil and Dry soil. Our method is more inclined to capture this crucial information on climatic conditions.

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TABLE IX
ABLATION STUDY OF NETWORK ARCHITECTURE

No.	CNN	Transformer	BFIN			Houston	Pavia	SH2HZ	NC
			GDR	SSCA	GATE				
1	✓	×	×	×	×	73.20±3.74	68.65±2.15	79.76±6.45	66.00±9.69
2	×	✓	×	×	×	74.09±1.38	79.02±3.64	82.65±2.48	69.78±9.14
3	✓	✓	×	×	✓	76.02±1.17	82.44±6.10	83.81±2.97	74.36±8.03
4	✓	✓	✓	×	×	74.13±1.80	81.05±4.90	82.76±3.14	70.95±6.49
5	✓	✓	✓	✓	×	75.04±1.26	82.03±5.62	83.65±2.73	74.50±8.36
6	✓	✓	✓	✓	✓	**76.73±1.26**	**85.36±1.59**	**86.60±2.06**	**82.82±3.62**

TABLE X
ABLATION STUDY

No.	CLS	TAL	Pseudo-Label	Houston	Pavia	SH2HZ	NC
1	✓	×	×	76.73±1.26 75.43±3.13 63.24±2.02	85.36±1.59 86.21±1.51 83.30±1.90	86.60±2.06 89.87±1.30 77.50±3.20	82.82±3.62 65.94±0.26 74.89±4.90
2	✓	✓	×	77.43±0.78 76.99±2.40 64.43±0.78	86.95±0.99 87.81±1.17 85.59±1.19	87.99±2.04 90.97±1.34 79.80±3.18	85.48±2.72 66.55±0.88 78.18±3.99
3	✓	✓	✓	**79.96±1.51 77.56±1.97 67.15±1.82**	**92.50±0.36 91.29±0.56 91.04±0.43**	**93.38±1.16 93.89±1.09 88.41±2.02**	**93.42±0.54 71.23±3.91 89.82±0.83**

D. Ablation Experiments

To verify the effectiveness of each component in the experiment, we conducted multiple ablation experiments. First, we divide Table IX into five parts. Among them, CNN or Transformer represents the use of a single network for feature extraction. BFIN is divided into three small parts for separate study, namely, GDR represents the bidirectional dynamic feature preservation mechanism, SSCA represents the spectral space cross-attention, and GATE represents the effective combination of GDR and SSCA through gating.

As can be seen from Table IX, the overall classification accuracy of using either CNN or Transformer alone on the four datasets is relatively low. The performance of the Transformer is slightly higher. This is because our Transformer is designed based on CNN, but it still discards some of the effective information of CNN or fails to combine the effective information of the two, so the overall performance is not optimistic. When we integrate CNN and Transformer with GATE, the classification performance has been partially improved. The classification accuracy rate of Houston mission reached 76%, and that of Pavia mission reached 82%, with more positive improvements. For SH2HZ and NC, they reached 83% and 74%, respectively. From this, it can be seen that GATE plays a crucial role in the entire module. In addition, we removed GATE and added GDR separately. At this point, the classification accuracy of the four datasets has all improved compared to when either CNN or Transformer is used alone. Then, we continue to add SSCA to further achieve the integration of cross-domain information. The addition of SSCA has increased the accuracy of NC tasks by approximately 4%, and the standard deviation has also tended to stabilize. This phenomenon is a good outcome of the DA task. Finally, let us continue to join GATE. At this point, GATE effectively integrated the GDR and SSCA components, and the classification accuracy rates of the four tasks reached 77%, 85%, 86%, and 82%, respectively. This is because the GATE controls the effective fusion of features. To sum up, embedding BFIN in a feature extractor that includes CNN and Transformer is a relatively novel design, and each component plays an important role.

Table X shows that we tested the effectiveness of loss L_{tal} and pseudo-labels. According to the data in the table, the classification effect of the first row as the baseline is the lowest. Then, the addition of L_{tal} achieves cross-domain feature-level alignment, which is more prominently reflected in AA and Kappa coefficients. For instance, the AA and Kappa coefficients of Houston mission have increased by approximately 1% compared to the first row. The three indicators of Pavia mission each improved by approximately 1%. The Kappa coefficient of the SH2HZ mission has increased by approximately 2%. Its OA and AA systems have also been improved. The OA and Kappa coefficients of the NC task have been significantly improved. Therefore, L_{tal} loss is relatively positive in DA tasks. Finally, we incorporate topology-aware multiview consistency pseudo-label optimization. The credibility of pseudo-labels was enhanced through strategies such as multiview voting and topological structure difference perception optimization for fine-tuning the model. This strategy has significant research significance in greatly enhancing the effectiveness of the model. In conclusion, these two modules also have significant significance in existence.

E. Parameter Sensitivity Analysis

Throughout the experiment, we selected six relatively important parameters for analysis to assess their sensitivity in four datasets. Fig. 12 shows the classification accuracy variation curves of parameter sensitivity on four datasets.

Learning Rate Analysis: In our experiment, we tested seven different learning rates to evaluate the impact of the experiment on four datasets. Fig. 12(a) shows the trend of classification accuracy varying with the size of the learning rate. In Houston task, the learning rate peaks at 0.1. As the learning rate decreases, the classification accuracy also shows a downward trend. Although it briefly rebounded at 0.0004, the performance remains relatively low. In Pavia task, as the learning

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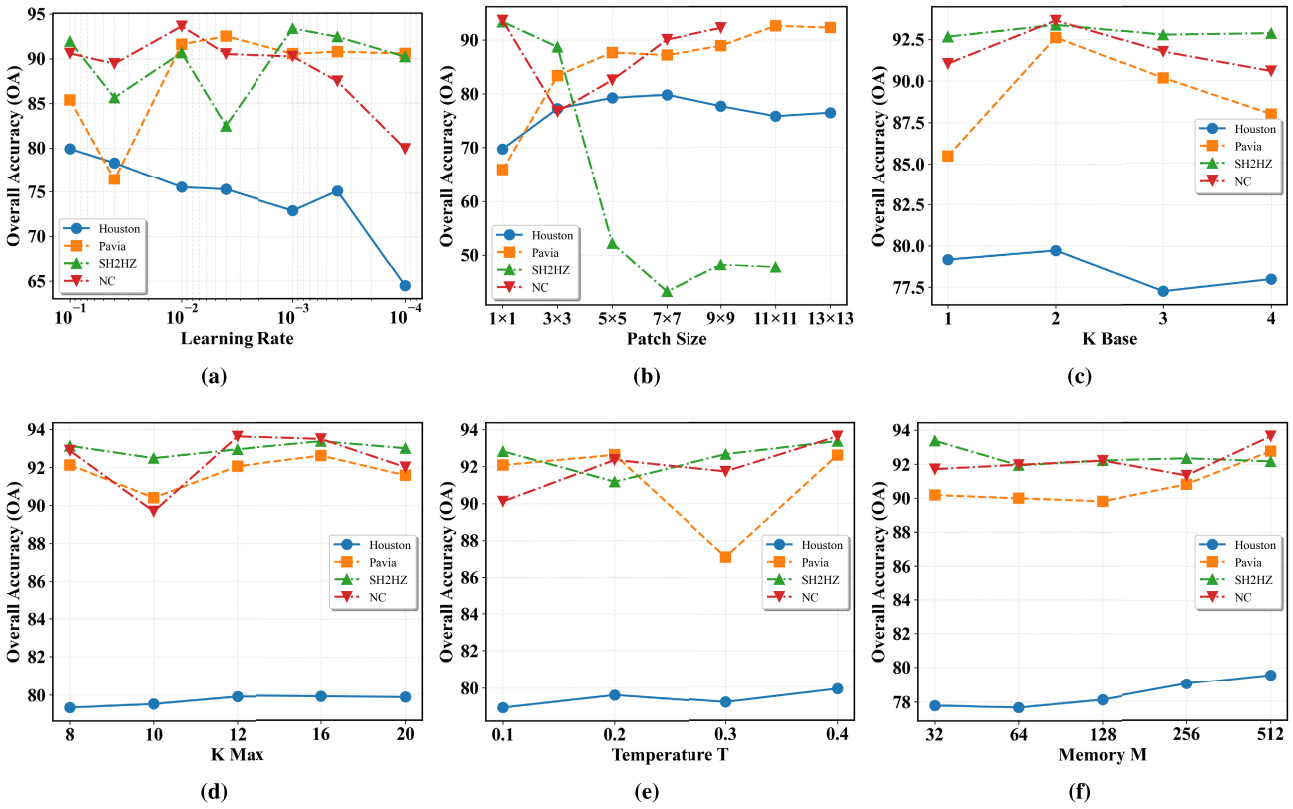


Fig. 12. (a) Learning rate. (b) Patch size. (c) K base. (d) K max. (e) Temperature T. (f) Memory M.

rate decreases, the classification accuracy first decreases and then increases, reaching the highest value at 0.004, and then shows a slow downward trend. Therefore, we have controlled the learning rate at 0.004. In the SH2HZ task, the variation of classification accuracy with the learning rate is rather complex. However, we captured 0.001 at these seven points, and the classification accuracy at this point serves as the highest value of the entire curve. In the NC task, as the learning rate decreases, the accuracy rate shows a trend of first decreasing, then increasing, and then decreasing again. The accuracy rate is the highest when the learning rate is 0.01. Therefore, we set the learning rate of the NC task to 0.01.

Analysis of Patch Size: We tested seven different image sizes on four datasets, respectively, to evaluate the impact on the experiment. Fig. 12(b) shows the variation curve of classification accuracy with image size. First, in Houston task, the classification accuracy showed an upward trend as the image size increased, reaching the highest value at 7×7. Then, as the size further increases, the classification accuracy gradually decreases. Second, in Pavia task, the classification accuracy gradually increased with the increase of image size before 11×11, and then showed a downward trend when it increased to 13×13. Therefore, we have controlled the size to 11×11. Finally, in the SH2HZ task, the curve of classification accuracy is rather special. The highest value was reached at 1×1, and then, a relatively significant downward trend began to emerge. Therefore, we set the image size of this task to 1×1. In the NC task, we mainly tested five different sizes. When the size of the NC task is set to 1×1, the classification accuracy is the highest.

We mainly tested four different parameter analysis topological alignment losses, namely, K_{Base}, K_{Max}, temperature, and memory. Fig. 12(c) tests four different K_{Base} values, whose function is to control the initial number of nearest neighbors in the early stage of training. That is, using a smaller K_{Base} in the early stage of training can focus on the most reliable local structure (avoiding noise interference). In Houston task, the classification accuracy shows a trend of first increasing and then decreasing with the increase of the K_{Base} value, and reaches the maximum value at $K_{Base} = 2$. The performance in Pavia mission is similar to that of Houston, but there is a significant trend of change. In the SH2HZ task, $K_{Base} = 2$ serves as the peak of classification accuracy, showing a slight downward trend at several other points. The K_{Base} curve variation of the NC task is similar to that of Pavia and SH2HZ tasks. When $K_{Base} = 2$, the classification accuracy is the highest.

Fig. 12(d) tests five different K_{Max} values, whose function is to limit the upper limit of dynamic adjustment and prevent the introduction of irrelevant noise samples, that is, samples with low similarity, due to excessive K_{Max}. In Houston task, the classification accuracy shows a trend of increasing first and then decreasing with the increase of the K_{Max} value. It reaches its peak at $K_{Max} = 12$, but the overall change curve is relatively gentle. Therefore, $K_{Max} = 12$ is selected as the important parameter value. In Pavia task, the classification accuracy shows a trend of first decreasing, then increasing, and then decreasing again as K_{Max} increases. The degree of change is slightly greater, and a peak appears at $K_{Max} = 16$. In the SH2HZ mission, the curve changes quite similar to those

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TABLE XI
PERIODIC VERIFICATION ANALYSIS

Verification	Houston	Pavia	SH2HZ	NC
0	78.71±0.78	90.56±0.98	91.93±0.86	91.92±1.47
1	79.11±0.94	91.25±1.08	92.88±0.81	92.07±1.61
5	**79.96±1.51**	**92.50±0.36**	93.04±0.64	92.24±0.92
10	78.79±1.31	92.06±0.45	**93.38±1.16**	**93.42±0.54**

TABLE XII
SSCA AND SINKHORN PARAMETER ANALYSIS EXPERIMENT

	Houston	Pavia	SH2HZ	NC
$\epsilon = 1$	79.14±0.75	91.22±0.26	90.30±1.88	82.21±7.11
$\epsilon = 0.1$	**79.96±1.51**	**92.50±0.36**	**93.38±1.16**	**93.42±0.54**
$\epsilon = 0.01$	77.79±1.29	91.61±0.26	91.55±0.42	88.44±5.61
$\epsilon = 0.001$	75.90±3.42	91.65±0.18	92.04±0.11	92.85±1.18
$N = 8$	74.70±0.73	89.91±0.49	89.70±1.34	92.30±0.92
$N = 16$	76.29±0.86	78.13±2.42	91.55±1.02	91.50±0.50
$N = 32$	**79.96±1.51**	**92.50±0.36**	**93.38±1.16**	**93.42±0.54**
$N = 64$	76.89±2.55	83.70±8.23	90.15±1.33	90.99±0.85

in Houston mission, but a peak occurs at $K_{Max} = 16$. The accuracy rate is the highest when the K_{Max} of the NC task is 12.

Fig. 12(e) tested four different temperatures to evaluate performance on four datasets. T is used to control the sharpening or smoothing degree of the similarity score. T will adaptively adjust according to the changes of K (where K can be either K_{Base} or K_{Max}), and be used for neighborhood selection and weight calculation. When K is small and T is low, it is used to focus on similar samples. When K increases, the temperature also rises moderately, and the model learns a more global structure. In Houston task, the classification accuracy shows a trend of first increasing, then decreasing, and then rising with the decrease of T, reaching the peak at $T = 0.4$. In Pavia mission, the trend of change is similar to that of Houston mission, but the degree of change is greater. In the SH2HZ task, there is a trend of first decreasing and then increasing, with $T = 0.4$ as the peak. The temperature value of the NC task shows an upward trend in accuracy as T increases. The accuracy rate is the highest when $T = 0.4$.

Fig. 12(f) tests five memory parameters of different sizes, which are used to store historical features and form an interactive topological structure with the current batch. It can be seen from the figure that the classification accuracy of Houston task shows a trend of first decreasing and then increasing with the increase of memory, reaching the maximum value when $M = 512$. The overall curve of Pavia shows an upward trend and reaches its peak at $M = 512$. On the contrary, SH2HZ reaches its maximum value when $M = 32$. Similarly, the value of M in the NC task is the same as that in Pavia task.

In TA-MVCPO, we tested verification cycles of 0, 1, 5, and 10 on four datasets, respectively, to select reasonable parameters to improve the authenticity of pseudo-labels. Table XI shows the classification accuracy verified in each period. When the verification period is 0, topological difference perception will no longer play a role, and the classification accuracy will also be the lowest. The best effect is achieved when the verification cycle of Houston and Pavia is 5, and that of SH2HZ and NC is 10. It can be seen from this that topological structure differences play an important role in the verification of source-domain labels and target-domain pseudo-labels.

Table XII presents the parameter analysis of the SSCA spectral subspace dimension N and the Sinkhorn regularization parameter ϵ. ϵ controls the smoothness of the Sinkhorn optimal transport matrix. By choosing the appropriate ϵ, the transport matrix becomes smoother and more accurate during the training process. For the regularization parameter ϵ, we tested four values of different magnitudes. The performance is the best on the four datasets when $\epsilon = 0.1$. This indicates that the regularization intensity ϵ during the training process is conducive to the precise matching of the model. For N, experiments show that when $N = 32$, the model achieves the best performance on all four datasets. When N is too small, the spectral grouping becomes too fine, resulting in an overly large spatial dimension and making it difficult to capture effective spectral correlations. When N is too large, spatial information is overly compressed, resulting in the loss of key local features.

F. Feature Visualization

This section aims to analyze the misclassification problem caused by class distribution shift through feature visualization by t-distributed stochastic neighbor embedding (t-SNE) random sampling. Specifically, we use ground-truth labels and model's predicted labels to visualize the features of the target domain on three datasets, respectively. Different categories are distinguished by using different colors. Fig. 13(a) and (b) shows the visualization of the characteristics of Houston dataset. It can be seen that although the performance is strong in most categories, there are many classification errors in category 6 in the figure. Fig. 13(c) and (d) shows the visualization of the features of Pavia dataset. It can be seen that category 3 in the figure appears in both categories 1 and 2, resulting in an incorrect classification of category 3. Fig. 13(e) and (f) shows the visualization of the features of the SH2HZ dataset. It can be seen that there are too many overlapping parts between categories 1 and 2 in the figure. Compared with the real label, category 2 has been wrongly classified as category 1. Fig. 13(g) and (h) shows the visualization of the features of the NC dataset. Category 4 is obviously mixed with other categories, which makes it difficult for our model to distinguish it during classification. Although our method has certain advantages in solving this classification problem, the misclassification problem caused by class distribution shift deserves further attention.

To verify the aggregation degree of TAGIN's samples, Fig. 13 also visualizes the feature distribution of the source domain and the target domain. As can be seen from Fig. 13(b), the distribution and aggregation of samples of the same category are relatively concentrated, and there are distant outliers in the target-domain categories 2–4. This indicates that there are difficult samples in this task that are prone to misclassification. Fig. 13(d) shows a significant aggregation phenomenon in the source- and target-domain samples of the same category. Although there are outliers in category 3, the model as a whole has learned good domain-invariant features. The aggregation

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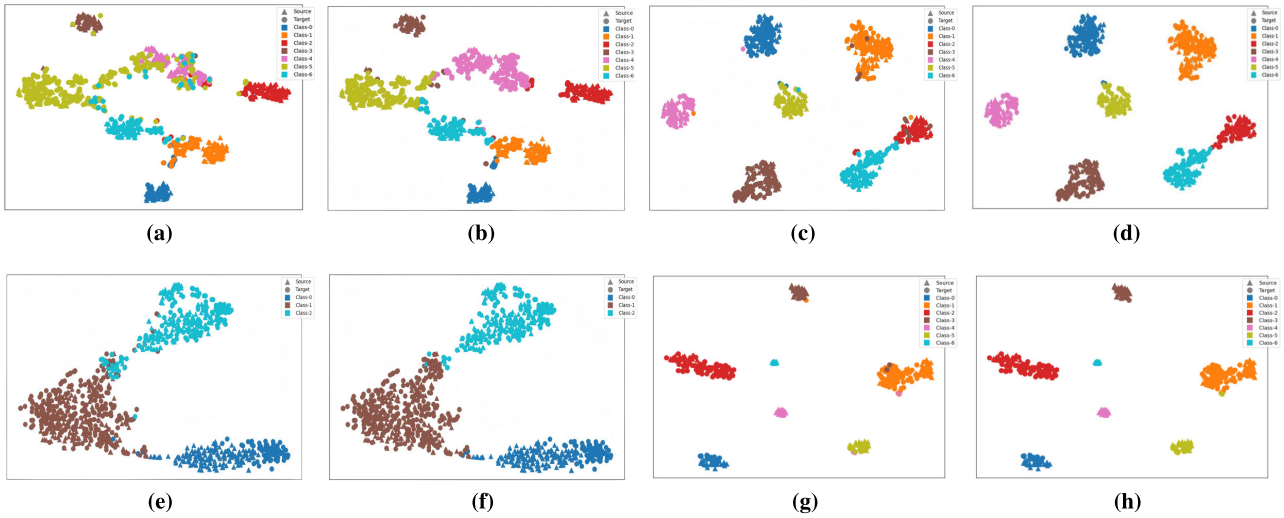


Fig. 13. Feature visualization on three datasets. Triangles represent the source domain, and circles represent the target domain. (a) Characteristics of the coloring of Houston ground-truth labels. (b) Characteristics of Houston pseudo-label coloring. (c) Characteristics of Pavia ground-truth labels. (d) Characteristics of Pavia pseudo-label coloring. (e) Characteristics of the coloring of SH2HZ ground-truth labels. (f) Characteristics of SH2HZ pseudo-label coloring. (g) Characteristics of the coloring of NC ground-truth labels. (h) Characteristics of NC pseudo-label coloring.

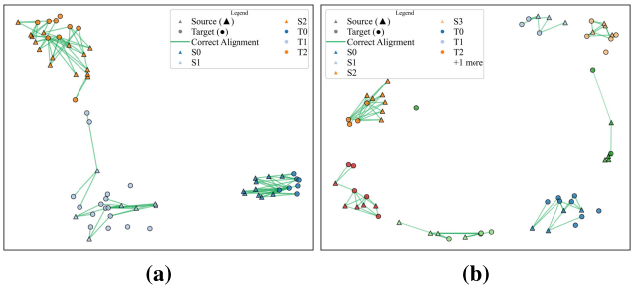


Fig. 14. Topological alignment loss for cross-domain alignment on (a) SH2HZ dataset and (b) Pavia dataset.

phenomenon in Fig. 13(f) and (h) is also quite obvious. All categories have shown an overlapping aggregation state in both the source domain and the target domain, and there is no obvious outlier phenomenon in the target domain. This indicates that the model has certain advantages in establishing domain-invariant features.

Fig. 14 presents the cross-domain alignment visualization results of the topological alignment loss on the SH2HZ dataset and Pavia dataset. These two datasets exhibit a certain degree of domain drift. As can be seen from the figure, this loss can effectively establish the topological relationship between the source domain and the target domain, forming a network structure across domains. This provides unique insights for analyzing the global information between cross-domain data and reducing the differences between domains. Therefore, our loss function can form a topological structure in the cross-domain mechanism based on the similarity differences, thereby reducing the differences between domains. This method provides a more comprehensive perspective for understanding the microscopic mechanism of unsupervised DA.

G. Computational Complexity

To compare the computational efficiency of various methods, the computational complexity of each method was evaluated by analyzing the training time, test time, the number of parameters, and the number of floating-point operations (FLOPS). Table XIII provides a relatively detailed analysis of the computational complexity. It can be seen from the data in the table that since the source- and target-domain data need to participate in the training simultaneously, the training time is relatively long. Moreover, the method proposed in this article requires feature mining and interaction between the source domain and the target domain in the BFIN module during training. Therefore, this part increases the training time, parameter quantity, and FLOPS of the model. By comparing with other methods, our method shows no significant differences in the number of model parameters, FLOPS, and training time. Although the testing time was slightly longer, the final performance result was superior to that of the existing domain adaptive methods.

To quantify and evaluate the computational overhead of the BFIN throughout model training, Table XIV counts the FLOPS and the proportion of model parameters of the BFIN in the entire model. BFIN is the core module of the entire model, with an overall complexity of approximately $O(B \cdot D^2)$. Here, B represents the input batch, and D represents the feature dimension. Its FLOPS range from 10.4% to 48.0% across the four datasets, and the number of model parameters is between 56.8% and 64.9%. Therefore, BFIN is an important source of computational overhead for the entire model. As an important cross-domain mechanism of BFIN, SSCA has FLOPS ranging from 4.5% to 20.9% across the four datasets, and the number of model parameters ranges from 47.3% to 54.0%. Although the computational complexity of these two components is relatively high, the ablation experiment has demonstrated their significant role in the entire model experiment. Furthermore, the multiscale and Sinkhorn modules are important parts of GDR in Table IX. Their statistics in terms of FLOPS and the number of model parameters are very small and negligible. GATE represents the effective combination of GDR and SSCA through gating. Other represents other important parts in the

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TABLE XIII
COMPUTATIONAL COMPLEXITY OF DIFFERENT METHODS ON DIFFERENT DATASETS

Methods	DANN	EDDMM	CDA	CLDA	TAADA	SCLUDA	MLUDA	MSDA	ASAL	Ours
Houston										
Training time(s)	36.58	1792.37	220.95	295.48	146.86	306.36	283.36	189.49	362.41	351.76
Testing time(s)	7.21	10.27	0.72	6.95	8.53	3.66	8.75	3.85	2.43	13.36
FLOPS	15873800	6555402	19848507	6569238	7678089	15225218	31389468	15320609	10376843	29523000
Params(M)	13.15	0.17	0.07	0.17	0.57	1.75	2.10	0.57	2.14	2.48
Pavia										
Training time(s)	36.36	308.43	216.65	203.38	141.01	384.85	384.85	428.09	263.72	602.29
Testing time(s)	7.06	7.32	0.76	2.59	4.78	4.78	3.92	3.3	1.84	11.66
FLOPS	23784392	11686202	23104507	11700038	50180201	80408762	161659781	50180201	7536437	96982812
Params(M)	13.32	0.18	0.09	0.18	0.66	3.92	1.87	0.66	2.15	2.57
SH2HZ										
Training time(s)	40.31	978.58	165.17	247.64	96.99	370.91	56.95	202.37	397.71	174.28
Testing time(s)	17.43	15.52	1.87	13.21	26.24	9.01	22.25	12.13	3.85	27.52
FLOPS	13634756	201402	30000000	4113983	29503089	1453894	3911343	10534812	6022562	6447343
Params(M)	13.62	0.20	0.11	0.20	0.81	2.09	2.45	0.81	1.46	2.71
NC										
Training time(s)	47.56	1479.39	310.71	124.66	156.82	837.48	123.50	210.16	404.59	315.35
Testing time(s)	22.31	58.76	1.55	13.17	13.65	16.83	9.02	16.09	7.60	30.26
FLOPS	16403720	1635002	33856507	1648838	29503089	80408762	33171593	14309825	4200437	7287718
Params(M)	13.85	0.21	0.13	0.22	0.81	1.87	2.55	0.92	2.18	2.83

TABLE XIV
COMPLEXITY ANALYSIS OF THE BFIN MODULE AND ITS SUBMODULES

	FLOPS(M)	Params(M)	FLOPS(%)	Params(%)
BFIN	3.05	1.61	10.4%-48.0%	56.8%-64.9%
SSCA	1.34	1.34	4.5%-20.9%	47.3%-54.0%
GDR	0.14	-	0.4%-2.1%	-
GATE	-	0.10	-	3.5%-4.0%
Other	1.57	0.16	5.3%-24.3%	5.6%-6.4%

BFIN. These two parts also account for a very small proportion of the total FLOPS and the number of parameters of the entire model.

Although TAGIN performs well in terms of accuracy, its modules, such as BFIN, cross-domain SSCA, and topological alignment mechanism, still have certain computational and memory overheads. Overall, TAGIN is more applicable to satellite remote sensing scenarios with high labeling costs. Through topological structure modeling, it can better distinguish land features with similar spectra but different spatial structures. However, in cases where computing power is limited, such as when applied to real-time HSI classification for UAVs, the SSCA mechanism in BFIN can control the number of parameters and computational complexity by simplifying the attention mechanism and adjusting the feature dimensions. During the inference stage, the depth and width of the interaction layer in BFIN can be dynamically adjusted according to the complexity of the input image to achieve adaptive allocation of computing resources. Additionally, for simple scenarios or samples with high confidence, the computation can be simplified, thereby further improving the inference speed.

V. CONCLUSION

In this article, we propose a BFIN embedded in the encoder for adaptively fusing the local features of CNN and the semantic information of Transformer. In the process of training, we designed a dynamic adaptive topological alignment loss. Through memory enhancement and dynamic parameter adjustment mechanisms, we achieved progressive optimization of cross-domain feature alignment. In addition, we designed a TA-MVCPO strategy to fine-tune the training model and solve the problem of obtaining high-confidence pseudo-labels. We conducted comparative and ablation experiments on four benchmark HSI datasets and verified the effectiveness of the method from multiple dimensions such as accuracy, model complexity, and parameter sensitivity.

Although the proposed method has significant advantages in accuracy, there is still great room for improvement in its complexity. Under conditions of limited computing power, such as deployment on drone platforms, more lightweight models are needed for real-time testing. Therefore, in the future, we will conduct further research on model lightweighting to enable it to be deployed on platforms with limited computing power.

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